

# Beyond FAIR assessment black boxes: ontology-based evaluations and the challenge of comparing FAIR scores

Alban Gaignard

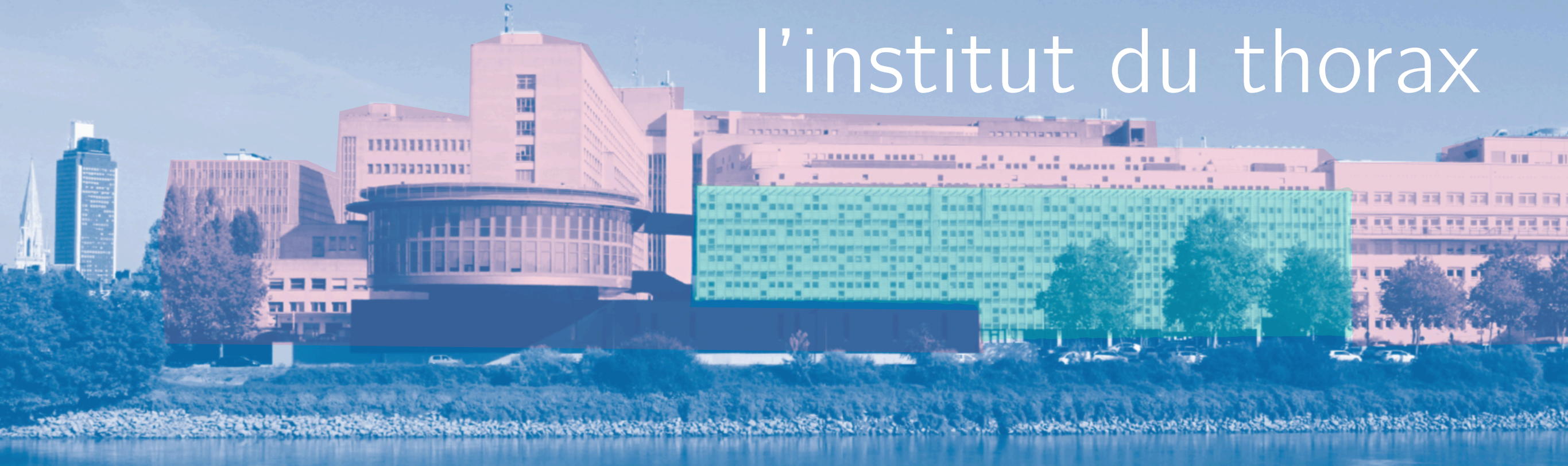
CNRS, institut du thorax, Nantes, France

October 20, 2024

CMLS 2025 - Poitiers, France

Context

# l'institut du thorax



Better understanding of cardiovascular and metabolic diseases

**Gene-function associations**

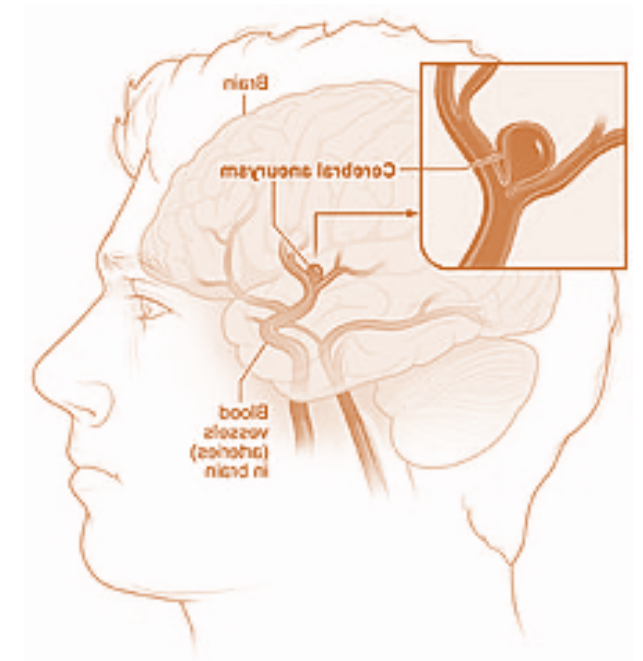
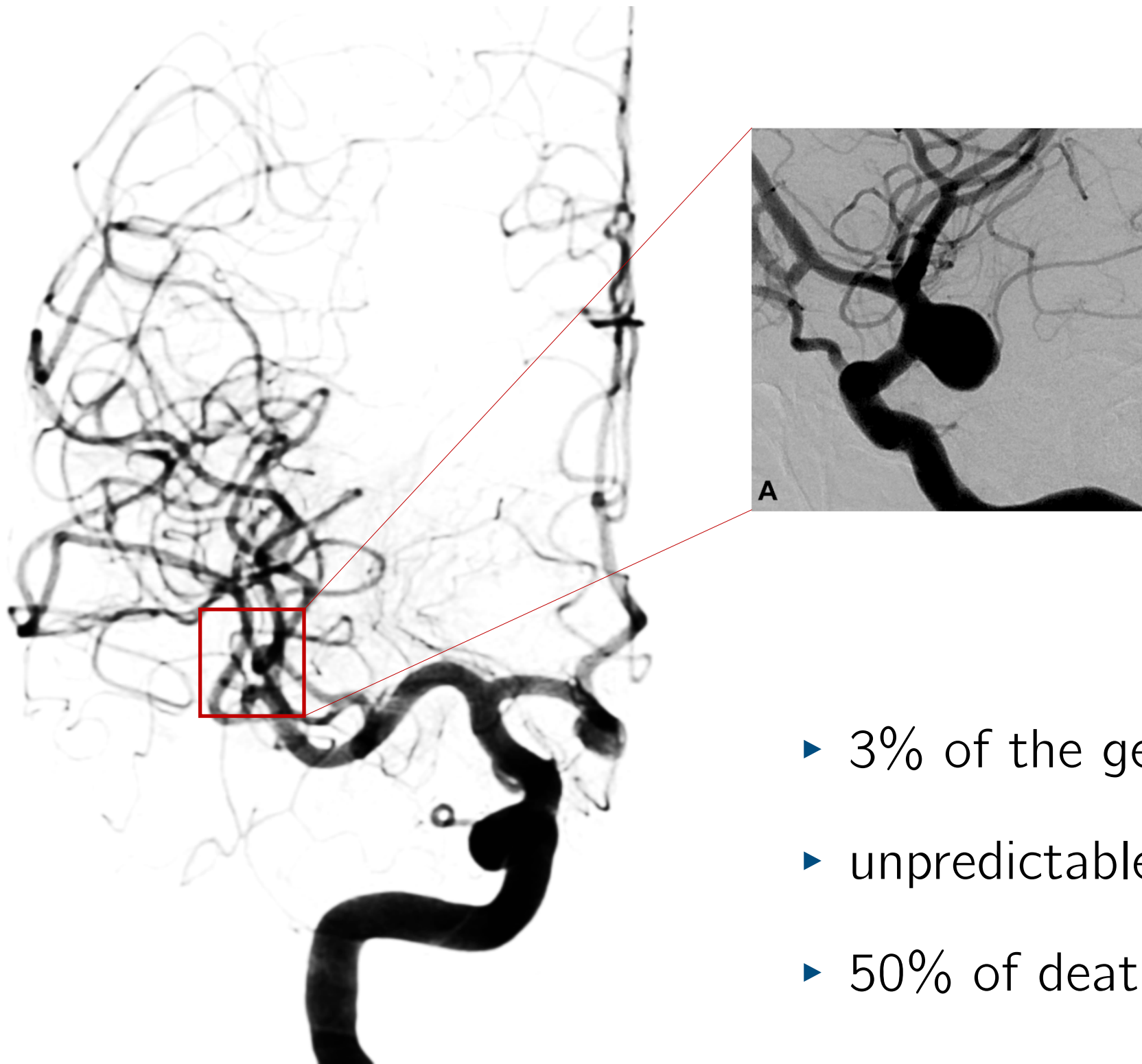
**Translational medicine**

Real proximity between the university hospital and the research lab

## Bioinformatics

- ▶ Massive production of genomic sequence & health data  
→ high performance computing
- ▶ Integration of multi-modal and multi-scale data
- ▶ Predictive models

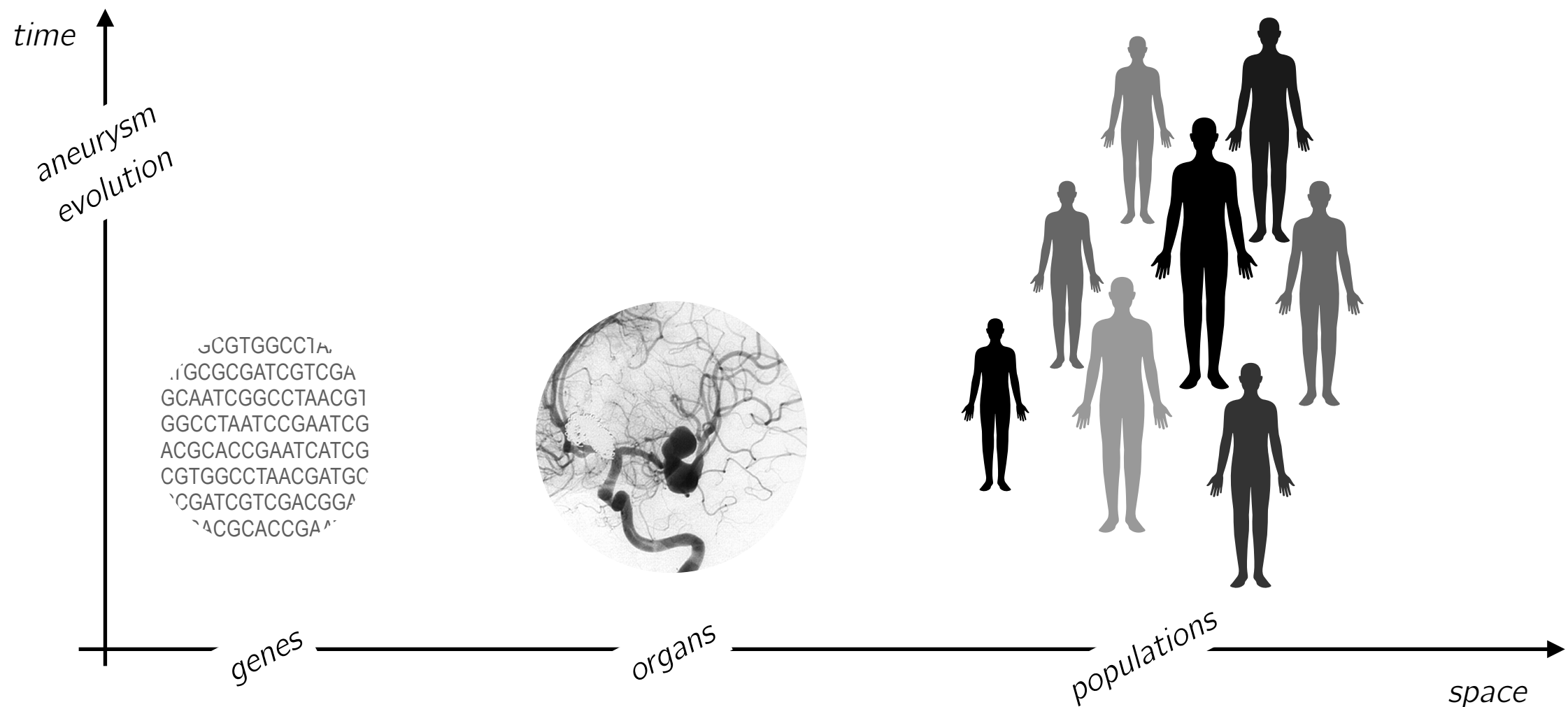
# Intracranial aneurysm (ICA)



- ▶ 3% of the general population
- ▶ unpredictable rupture
- ▶ 50% of death in case of rupture

# Multi-factorial disease → multi-scale data

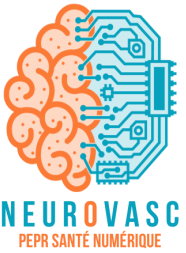
- ▶ Inter-disciplinary efforts needed for a better understanding of the pathology
- ▶ Specific data produced at very specific scales



# Neurovasc project



PROGRAMME  
DE RECHERCHE  
SANTÉ  
NUMÉRIQUE



- ▶ Neurovasc: a national programme funded for 4 years by the french research agency to build a **digital infrastructure** to manage and exploit **intracranial aneurysm data**
  - 3 Research Institutes (Inria, Inserm, IMT Atlantique)
  - 2 Clinical Research Teams (Brest & Nantes academic hospitals)
  - 3 Universities (Bordeaux, Paris-Saclay, Nantes)

## WP 1: A model of interoperable infrastructure between research and healthcare

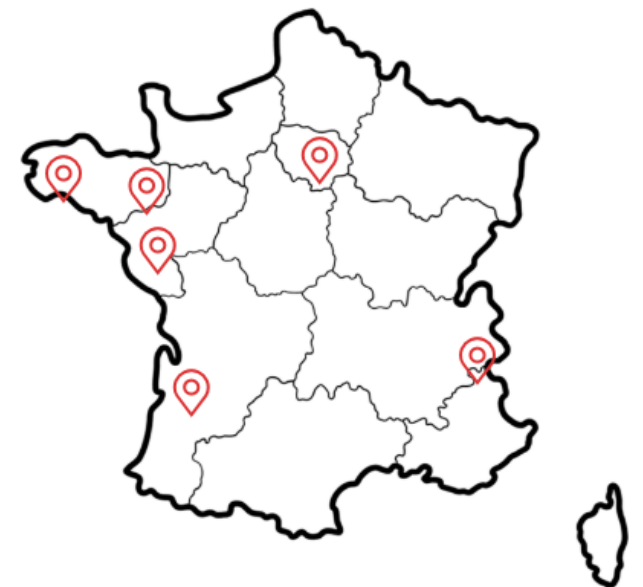
- Task 1.1. Managing clinical data
- Task 1.2. Managing imaging data
- Task 1.3. Managing genetic data

## WP 2: Interoperable datasets and predictive models for ICA diagnosis and outcomes

- Task 2.1: FAIR genomic data demonstrator
- Task 2.2: Mining healthcare circuits following ICA diagnosis
- Task 2.3: A non-additive model for global genetic-risk prediction

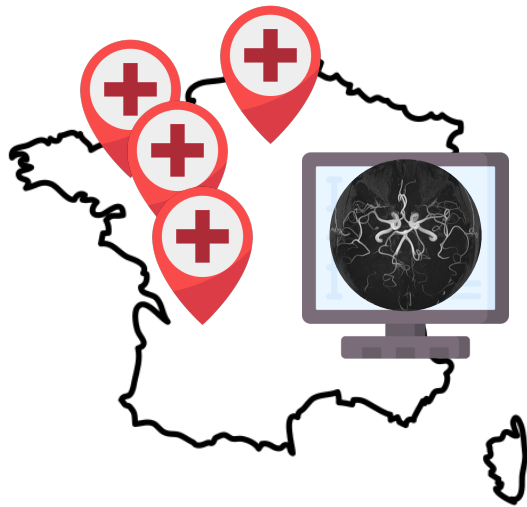
## WP 3: Proof-of-Concept studies with digital companions

- Task 3.1: Accompanying patients with diagnosed unruptured ICA
- Task 3.2: Post-stroke prevention through mobile applications and digital monitoring



# Data integration & sharing challenges

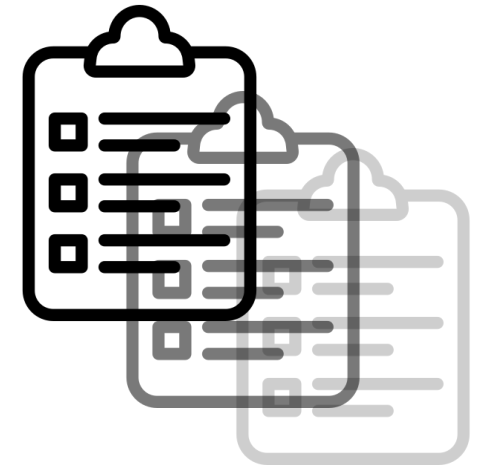
❶ How to **collect high-quality medical images** from multiple hospitals/MRIs ?



❷ How to **interlink and query multi-modal and multi-scale data** while preserving privacy constraints ?



❸ How to mine and **model patient trajectories** from EHR data ? can we predict clinical outcomes ?

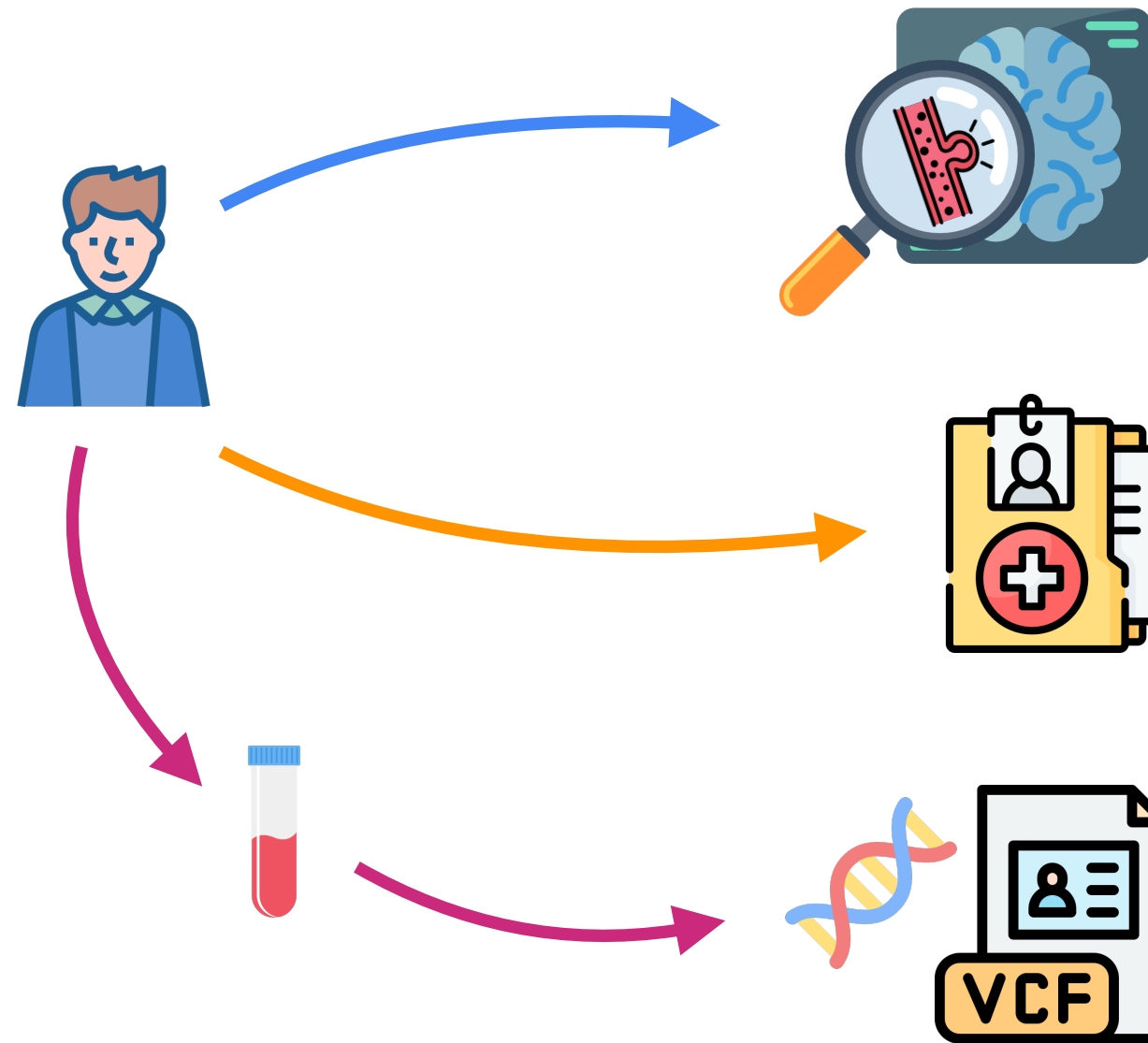


**We need FAIR data !**

# FAIRifying clinical & genomic data



Alexandrina  
Bodrug



Neuro-vascular imaging / tissues ?

- **UBERON**
- **NCIT**

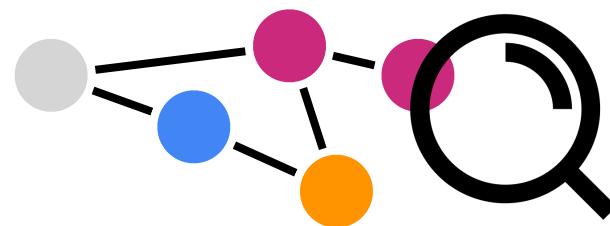
Clinical data / phenotypes ?

- **SPHN**
- **HPO**
- **DUO**

Genomic data ?

- **FALDO**
- **SO / GENO**
- **SIO**

A clinical and genomic  
intracranial aneurysm  
knowledge graph



to find & exchange  
phenoytypes/variants with  
reference terminologies !



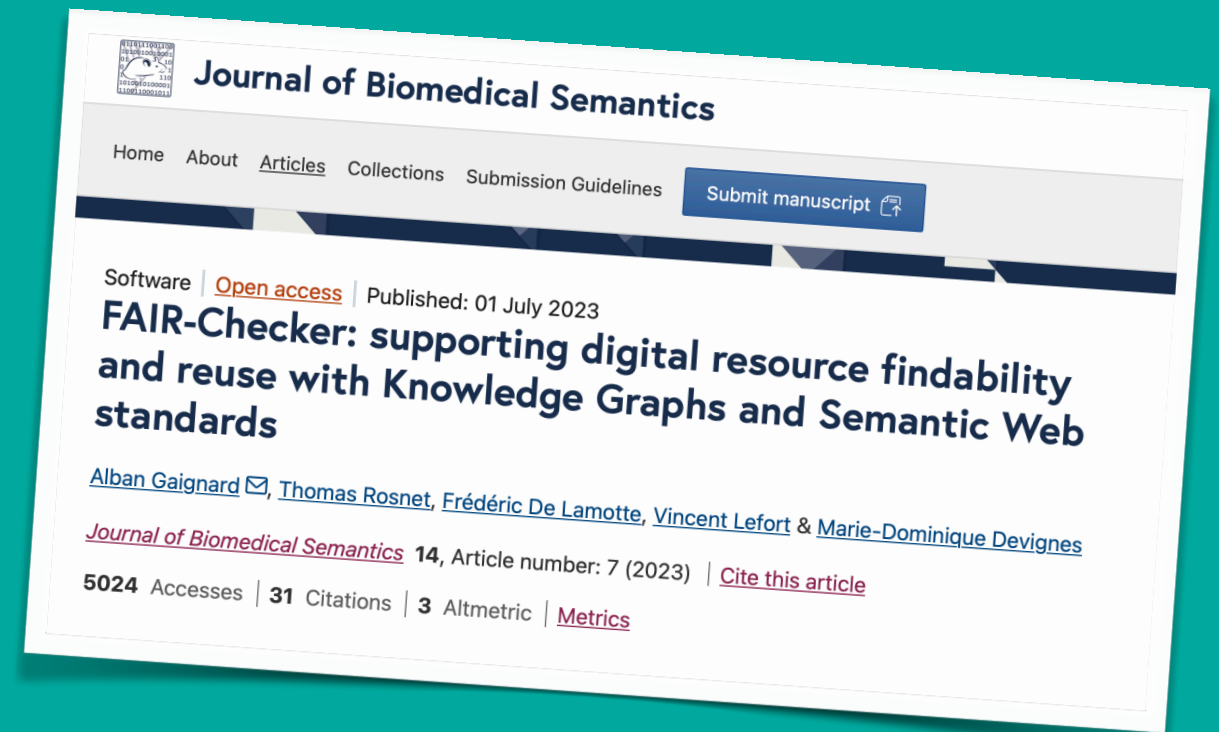
Marie-Dominique Devignes



Thomas Rosnet



Frédéric de Lamotte



# How can we evaluate that research data are FAIR enough ?

# FAIR principles require tooling



Australian Research Data Commons

*Web URIs*

*HTTP, RDF/SPARQL*

*Domain ontologies*

## FAIR principles

- ▶ critical for open & reproducible sciences
- ▶ result in many guidelines
- ▶ technology agnostic guidelines

How to implement the principles ...  
... and go beyond checklists ?

**Resource provider  
and developers  
need **help** and **tooling**.**

<https://www.go-fair.org/fair-principles>

<https://www.nature.com/articles/sdata201618>

# Usage scenarios



**Data producers**

**Make**



- Dataset
- Training
- Software tools
- etc ...

**discoverable !**

**Where to publish ?**

**Which registry ?**

**Does it provide metadata ?**

**Is it enough to be FAIR ?**



**Registry developers**

**Make**



- Dataverse
- Bio.tools
- Zenodo,  
Pubmed ...

**more FAIR !**

**Improve metadata quality ?**

**Community specific  
standards ?**

**Which technology ?**

# Why a (nother) tool ?

## Assumptions

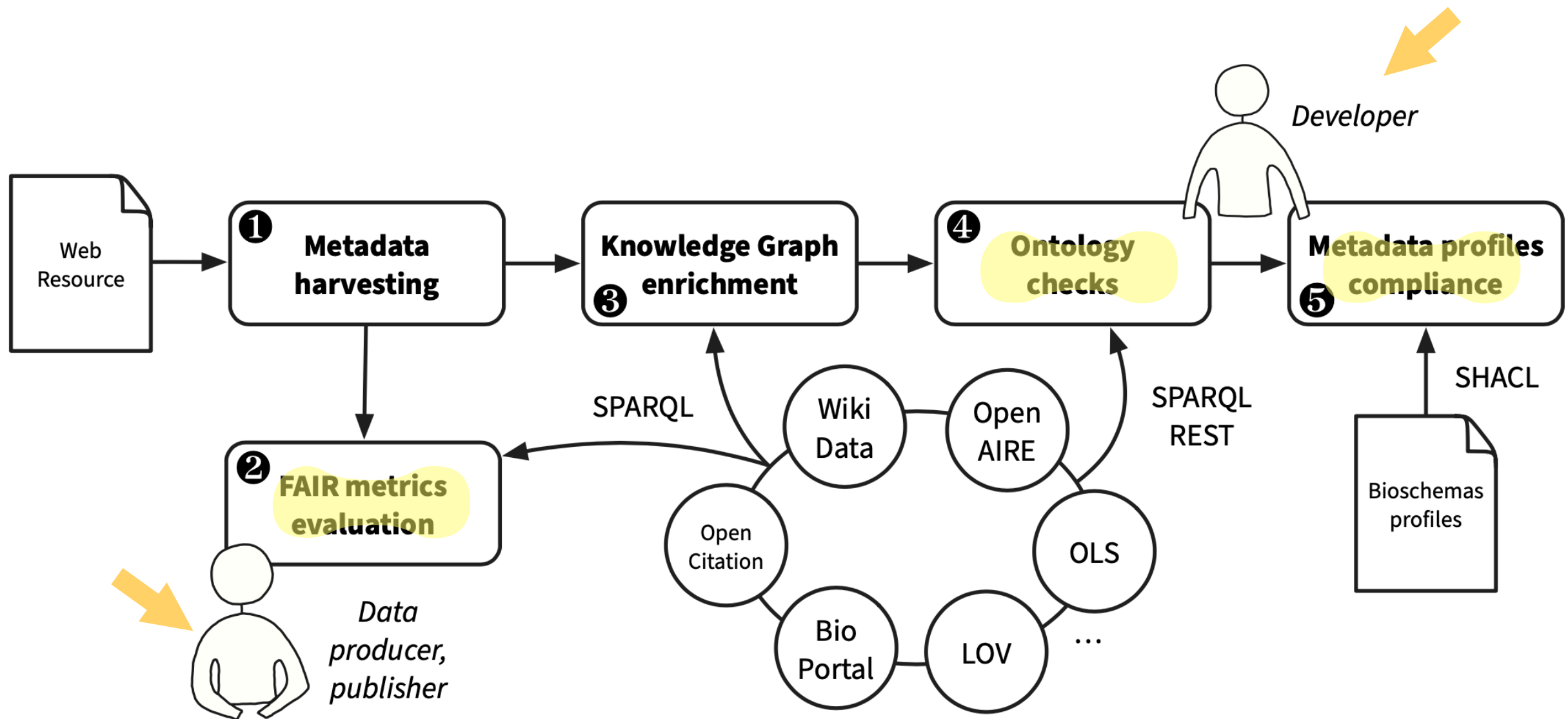
- ▶ "Linked Data" and Semantic Web technologies are key in most of the FAIR principles (especially F, I, and R)  
... but technical skills are needed.

## Objectives

- ▶ Provide a web interface for resource providers to **evaluate FAIR metrics** and **make progress on FAIRification** (iterative testing)
- ▶ Provide additional tools for developers Leverage semantic web technologies (RDF, SPARQL, SHACL) to **enhance the quality of metadata**
- ▶ Share implementations, and contribute to larger FAIR evaluation initiatives



# Approach



**Figure 1** Gathering, enriching and analyzing semantic web annotations in line with FAIR principles.

# ① Checking metadata

# Checking metadata



Semantic web technologies

- RDF
- SPARQL

<https://fairplus.github.io/the-fair-cookbook/content/home.html>

<https://rdmkit.elixir-europe.org>

# What is evaluated ?

| Findability<br>F1B, F2                                                                                                                                    | Accessibility<br>A1.2                            | Reuse (licenses)<br>R1.1                                                                                                | Reuse (provenance)<br>R1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dct:identifier<br>schema:identifier<br>dct:title<br>dct:description<br>dcat:accessURL<br>dcat:downloadURL<br>dcat:endpointDescription<br>dcat:endpointURL | odrl:hasPolicy<br>dct:rights<br>dct:accessRights | schema:license<br>dct:license<br>doap:license<br>dbo:license<br>cc:license<br>xhv:license<br>sto:license<br>nie:license | prov:wasGeneratedBy<br>prov:wasDerivedFrom<br>prov:wasAttributedTo<br>prov:used<br>prov:wasInformedBy<br>prov:wasAssociatedWith<br>prov:startedAtTime<br>prov:endedAtTime<br>dct:hasVersion<br>dct:isVersionOf<br>dct:creator<br>dct:contributor<br>dct:publisher<br>pav:hasVersion<br>pav:version<br>pav:hasCurrentVersion<br>pav:createdBy<br>pav:authoredBy<br>pav:retrievedFrom<br>pav:importedFrom<br>pav:createdWith<br>pav:retrievedBy<br>pav:importedBy<br>pav:curatedBy<br>pav:createdAt<br>pav:previousVersion<br>schema:creator<br>schema:author<br>schema:publisher<br>schema:provider<br>schema:funder |

DC-Terms  
DCAT  
Schema.org

+ ODRL

+ DOAP,  
DBO, CC ...

+ PROV-O, PAV

**Table 2** Summary of the selected ontology properties relevant to assess three specific FAIR principles in *FAIR-Checker*

**Resource identifier (URL/DOI)**

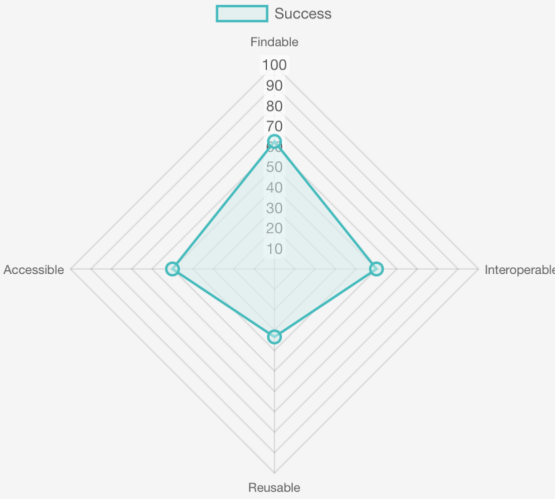
   [All metrics](#)

Valid URL/DOI - The input contains the following DOIs that you can also test:  10.3205/09dgnc137



[Dataset Dataverse](#) [Workflow](#) [Publication Datacite](#) [Dataset](#) [Tool](#)

# FAIR compliance



## Share your results

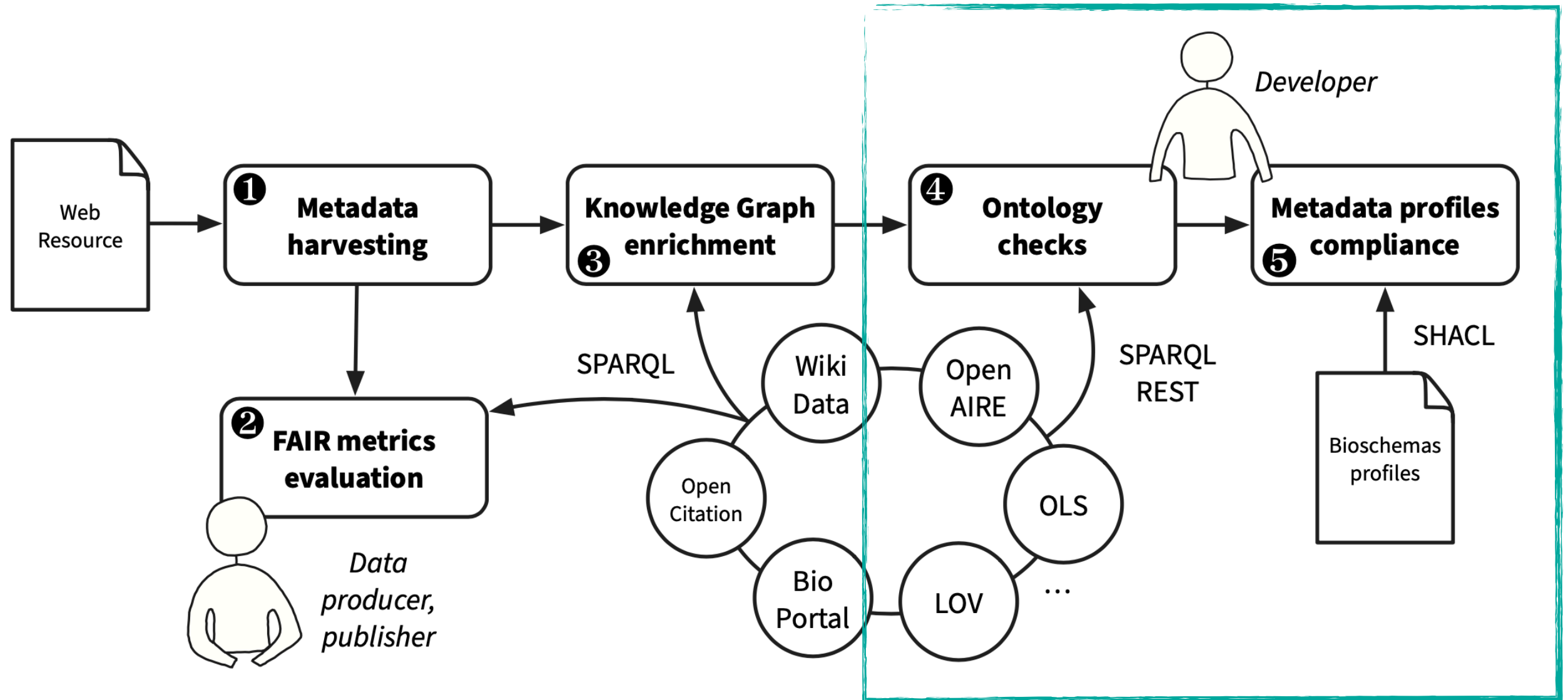
FAIR assessment 50.00 %

|                                                |                       |                         |                                                                                                                                                                                                |                   |
|------------------------------------------------|-----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| F2B: Shared vocabularies for metadata          | <a href="#">Check</a> | FAIR principle F2B 2/2  |                                                                                                                                                                                                | <a href="#">i</a> |
| A1.1: Open resolution protocol                 | <a href="#">Check</a> | FAIR principle A1.1 2/2 |                                                                                                                                                                                                | <a href="#">i</a> |
| A1.2: Authorisation procedure or access rights | <a href="#">Check</a> | FAIR principle A1.2 0/2 | You should describe the access policy in metadata by using at least one of the                                                                                                                 | <a href="#">i</a> |
|                                                |                       |                         | <a href="#">Read more</a>                                                                                                                                                                      |                   |
| I1: Machine readable format                    | <a href="#">Check</a> | FAIR principle I1 1/2   | You should provide discoverability oriented metadata with one of the following properties: dct:title dct:description dcat:accessURL dcat:downloadURL dcat:endpointDescription dcat:endpointURL | <a href="#">i</a> |
|                                                |                       |                         | <a href="#">Read less</a>                                                                                                                                                                      |                   |
| I2: Use shared ontologies                      | <a href="#">Check</a> | FAIR principle I2 2/2   |                                                                                                                                                                                                | <a href="#">i</a> |
| I3: External links                             | <a href="#">Check</a> | FAIR principle I3 0/2   | You should enrich your metadata with more diversified external links. Here we did not                                                                                                          | <a href="#">i</a> |
|                                                |                       |                         | <a href="#">Read more</a>                                                                                                                                                                      |                   |
| R1.1: Metadata includes license                | <a href="#">Check</a> | FAIR principle R1.1 0/2 | You should include information about licence in your metadata using one of the                                                                                                                 | <a href="#">i</a> |
|                                                |                       |                         | <a href="#">Read more</a>                                                                                                                                                                      |                   |

1. Provide resource URL
2. Retrieve metadata
3. Evaluate relevant metadata fields (ontology classes and properties)
4. Propose **relevant recommendations** from FAIR Cookbook and RDMkit
5. Improve metadata and **try** again ...

## ② Inspecting metadata

# Approach



**Figure 1** Gathering, enriching and analyzing semantic web annotations in line with FAIR principles.

# schema.org



## Full Hierarchy

Schema.org is defined as two hierarchies: one for textual property values, and one for the things that they describe.

This is the main schema.org hierarchy: a collection of types (or "classes"), each of which has one or more parent types. Although a type may have more than one super-type, here we show each type in one branch of the tree only. There is also a parallel hierarchy for **data types**.

## Types:

Close hierarchy / Open hierarchy

- Thing
  - ▶ Action +
  - ▶ BioChemEntity +
  - ▶ CreativeWork +
  - ▶ Event +
  - ▶ Intangible +
  - ▶ MedicalEntity +
  - ▶ Organization +
  - ▶ Person +
  - ▶ Place +
  - Product
    - DietarySupplement
    - Drug
    - IndividualProduct
    - ProductCollection
    - ProductGroup

- ▶ General purpose **lightweight** ontology
- ▶ Aimed at **annotating web pages**
- ▶ Targetting **FINDABILITY**
- ▶ Originating from major search engines



# Schema.org is massively adopted

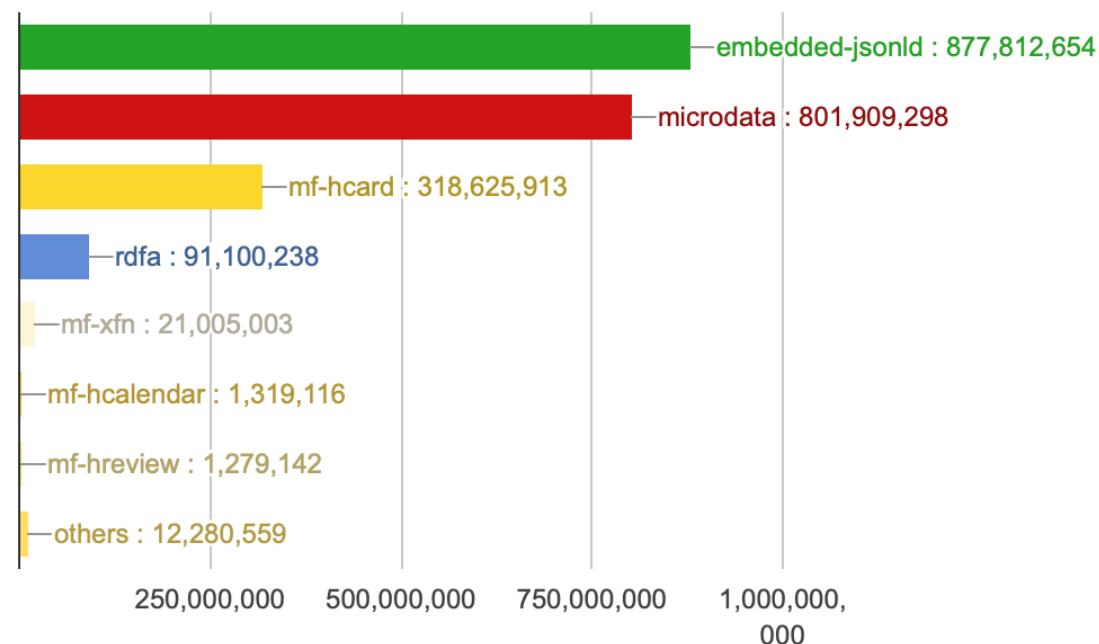
## Web Data Commons

Extracting Structured Data from the Common Crawl



|                        |                |              |
|------------------------|----------------|--------------|
| Crawl Date             | October 2022   |              |
| Total Data             | 82.71 Terabyte | (compressed) |
| Parsed HTML URLs       | 3,048,746,652  |              |
| URLs with Triples      | 1,518,609,988  |              |
| Domains in Crawl       | 33,820,102     |              |
| Domains with Triples   | 14,235,035     |              |
| Typed Entities         | 19,072,628,514 |              |
| Triples                | 86,462,816,435 |              |
| Size of Extracted Data | 1.6 Terabyte   | (compressed) |

### URLs with Triples



## Top Domains by Extracted Triples

1. [blogspot.com](https://www.blogspot.com) (879,564,145 triples)
2. [wordpress.com](https://www.wordpress.com) (458,770,038 triples)
3. [wikipedia.org](https://www.wikipedia.org) (190,087,065 triples)
4. [yummly.com](https://www.yummly.com) (87,112,540 triples)
5. [hotels.com](https://www.hotels.com) (81,991,039 triples)
6. [boohoo.com](https://www.boohoo.com) (79,884,394 triples)
7. [kayak.com](https://www.kayak.com) (77,623,248 triples)
8. [google.com](https://www.google.com) (73,729,078 triples)
9. [yahoo.com](https://www.yahoo.com) (65,317,838 triples)
10. [southleedslife.com](https://www.southleedslife.com) (63,758,451 triples)
11. [indiatimes.com](https://www.indiatimes.com) (58,899,559 triples)
12. [freepik.com](https://www.freepik.com) (56,124,447 triples)
13. [airbnb.com](https://www.airbnb.com) (51,964,983 triples)
14. [pinterest.com](https://www.pinterest.com) (47,251,484 triples)
15. [soundcloud.com](https://www.soundcloud.com) (45,745,317 triples)
16. [apple.com](https://www.apple.com) (42,410,414 triples)
17. [hostadvice.com](https://www.hostadvice.com) (42,309,867 triples)
18. [elpais.com](https://www.elpais.com) (42,136,136 triples)
19. [vsemayki.ru](https://www.vsemayki.ru) (38,167,517 triples)
20. [smugmug.com](https://www.smugmug.com) (38,031,434 triples)
21. [More](#)

Schema.org for  
Life-Science resources ?

# 37 ± Life Science profiles

| Name                                                                            | Group                                    | Use Cases                                                                           | Cross Walk                                                                            | Task & Issues                                                                         | Examples                                                                              | Live Deploys                                                                          |
|---------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <u><a href="#">ChemicalSubstance</a></u><br>(v0.4-RELEASE)<br>07 April 2020     | <u><a href="#">Chemicals</a></u>         |    |    |    |    |    |
| <u><a href="#">ComputationalTool</a></u><br>(v1.0-RELEASE)<br>11 October 2021   | <u><a href="#">Tools</a></u>             |    |    |    |    |    |
| <u><a href="#">ComputationalWorkflow</a></u><br>(v1.0-RELEASE)<br>09 March 2021 | <u><a href="#">Workflow</a></u>          |    |    |    |    |    |
| <u><a href="#">DataCatalog</a></u><br>(v0.3-RELEASE-2019_07_01)<br>01 July 2019 | <u><a href="#">Data Repositories</a></u> |    |    |    |    |    |
| <u><a href="#">Dataset</a></u><br>(v0.3-RELEASE-2019_06_14)<br>14 June 2019     | <u><a href="#">Datasets</a></u>          |  |   |  |  |   |
| <u><a href="#">FormalParameter</a></u><br>(v1.0-RELEASE)<br>09 March 2021       | <u><a href="#">Workflow</a></u>          |  |  |  |  |  |
| <u><a href="#">Gene</a></u><br>(v1.0-RELEASE)<br>07 April 2021                  | <u><a href="#">Genes</a></u>             |  |  |  |  |  |
| <u><a href="#">MolecularEntity</a></u><br>(v0.5-RELEASE)<br>07 April 2020       | <u><a href="#">Chemicals</a></u>         |  |  |  |  |  |
| <u><a href="#">Protein</a></u><br>(v0.11-RELEASE)<br>07 April 2020              | <u><a href="#">Proteins</a></u>          |  |  |  |  |  |
| <u><a href="#">Sample</a></u><br>(v0.2-RELEASE-2018_11_10)<br>10 November 2018  | <u><a href="#">Samples</a></u>           |  |  |  |  |  |
| <u><a href="#">Taxon</a></u><br>(v0.6-RELEASE)<br>07 April 2020                 | <u><a href="#">Biodiversity</a></u>      |  |  |  |  |  |

- ▶ different usage of schema.org for life sciences
- ▶ Communities agree on **minimal**/**recommended**/**optional** annotation

# Metadata completeness

## R1.3: (Meta)data meet domain-relevant community standards

| Marginality: Recommended.     |                                      |                                                                                                                                                                                                                                                                                                                                          |      |                                                                                                                                                                             |
|-------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>applicationCategory</u>    | <u>Text</u><br><u>URL</u>            | <p><b>Schema:</b><br/>Type of software application, e.g. 'Game, Multimedia'.</p> <p><b>Bioschemas:</b><br/>Type of tool e.g. Command-line tool, Web application etc. <b>Note:</b> Bioschemas have changed <u>URL</u> to <u>Text</u> in the Expected Types. This will be reverted once Bio.Tools provides stable URIs for tool types.</p> | MANY | <p>Please use terms from the 'Tool type' table in the <u>biotools documentation</u></p>  |
| <u>applicationSubCategory</u> | <u>Text</u><br><u>URL</u>            | <p><b>Schema:</b><br/>Subcategory of the application, e.g. 'Arcade Game'.</p> <p><b>Bioschemas:</b><br/>Use an <u>EDAM:Topic</u> to describe the category of application</p>                                                                                                                                                             | MANY | <p><u>EDAM:Topic</u></p>                                                                 |
| <u>author</u>                 | <u>Organization</u><br><u>Person</u> | <p><b>Schema:</b><br/>The author of this content or rating. Please note that author is special in that HTML 5 provides a special mechanism for indicating authorship via the rel tag. That is equivalent to this and may be used interchangeably.</p>                                                                                    | MANY |                                                                                        |
| <u>citation</u>               | <u>CreativeWork</u><br><u>IRI</u>    | <p><b>Schema:</b><br/>A citation or reference to another creative work, such as another publication</p>                                                                                                                                                                                                                                  | MANY |                                                                                        |

```
ex:myTool    rdf:type    schema:SoftwareApplication, prov:SoftwareAgent ;
schema:description "This tool does ... " ;
schema:license <https://spdx.org/licenses/MIT.html> ;
schema:codeRepository <http://github.com/...> .
```

# SHACL constraint template

```
1      ns:{{shape_name}} a sh:NodeShape ;
2      {% for c in target_classes %}
3      sh:targetClass {{c}} ;
4      {% endfor %}
5
6      {% for min_prop in min_props %}
7      sh:property [
8          sh:path {{min_prop}} ;
9          sh:minCount 1 ;
10         sh:severity sh:Violation
11     ] ;
12     {% endfor %}
13
14     {% for rec_prop in rec_props %}
15     sh:property [
16         sh:path {{rec_prop}} ;
17         sh:minCount 1 ;
18         sh:severity sh:Warning
19     ] ;
20     {% endfor %}
21     .
```

# Metadata completeness

R1.3: (Meta)data meet domain-relevant community standards

Validation of Bioschemas profiles:

- **rank missing** metadata
- developer **focus** on **minimal** metadata first

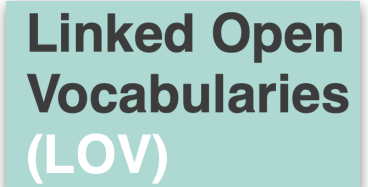
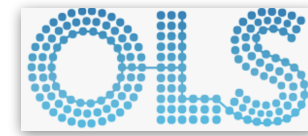
Check BioSchemas

`https://workflowhub.eu/workflows/18?version=1` has type `http://schema.org/ComputationalWorkflow`

Using `https://bioschemas.org/profiles/ComputationalWorkflow/1.0-RELEASE` for validation, specified from the `dct:conformsTo` property.

| Required missing properties                                    | Improvements                                                                   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>https://schema.org/input</code> <b>must be</b> provided  | <code>https://schema.org/citation</code> <b>should be</b> provided             |
| <code>https://schema.org/output</code> <b>must be</b> provided | <code>https://schema.org/contributor</code> <b>should be</b> provided          |
|                                                                | <code>https://schema.org/creativeWorkStatus</code> <b>should be</b> provided   |
|                                                                | <code>https://schema.org/documentation</code> <b>should be</b> provided        |
|                                                                | <code>https://schema.org/funding</code> <b>should be</b> provided              |
|                                                                | <code>https://schema.org/hasPart</code> <b>should be</b> provided              |
|                                                                | <code>https://schema.org/isBasedOn</code> <b>should be</b> provided            |
|                                                                | <code>https://schema.org/maintainer</code> <b>should be</b> provided           |
|                                                                | <code>https://schema.org/publisher</code> <b>should be</b> provided            |
|                                                                | <code>https://schema.org/runtimePlatform</code> <b>should be</b> provided      |
|                                                                | <code>https://schema.org/softwareRequirements</code> <b>should be</b> provided |
|                                                                | <code>https://schema.org/targetProduct</code> <b>should be</b> provided        |

# Reuse of ontologies



F4: (Meta)data are registered or indexed in a searchable resource

I2: (Meta)data use vocabularies that follow the FAIR principles

R1.3: (Meta)data meet domain-relevant community standards

## Step 3: Metadata quality checks

Controlled vocabularies

Bioschemas

We now have a Knowledge Graph grounded to ontology concepts (classes) and relations (properties). Are these classes and properties already known in reference ontology registries such as [LOV](#), [OLS](#) or [BioPortal](#) ?

Check Vocabularies

Congratulations ! All Classes and Properties are referenced in one or more of the registries checked !

### Classes

|                                                                             |     |     |           |
|-----------------------------------------------------------------------------|-----|-----|-----------|
| <a href="http://schema.org/DataDownload">http://schema.org/DataDownload</a> | OLS | LOV | BioPortal |
| <a href="http://schema.org/Organization">http://schema.org/Organization</a> | OLS | LOV | BioPortal |
| <a href="http://schema.org/Person">http://schema.org/Person</a>             | OLS | LOV | BioPortal |
| <a href="https://schema.org/Dataset">https://schema.org/Dataset</a>         | OLS | LOV | BioPortal |

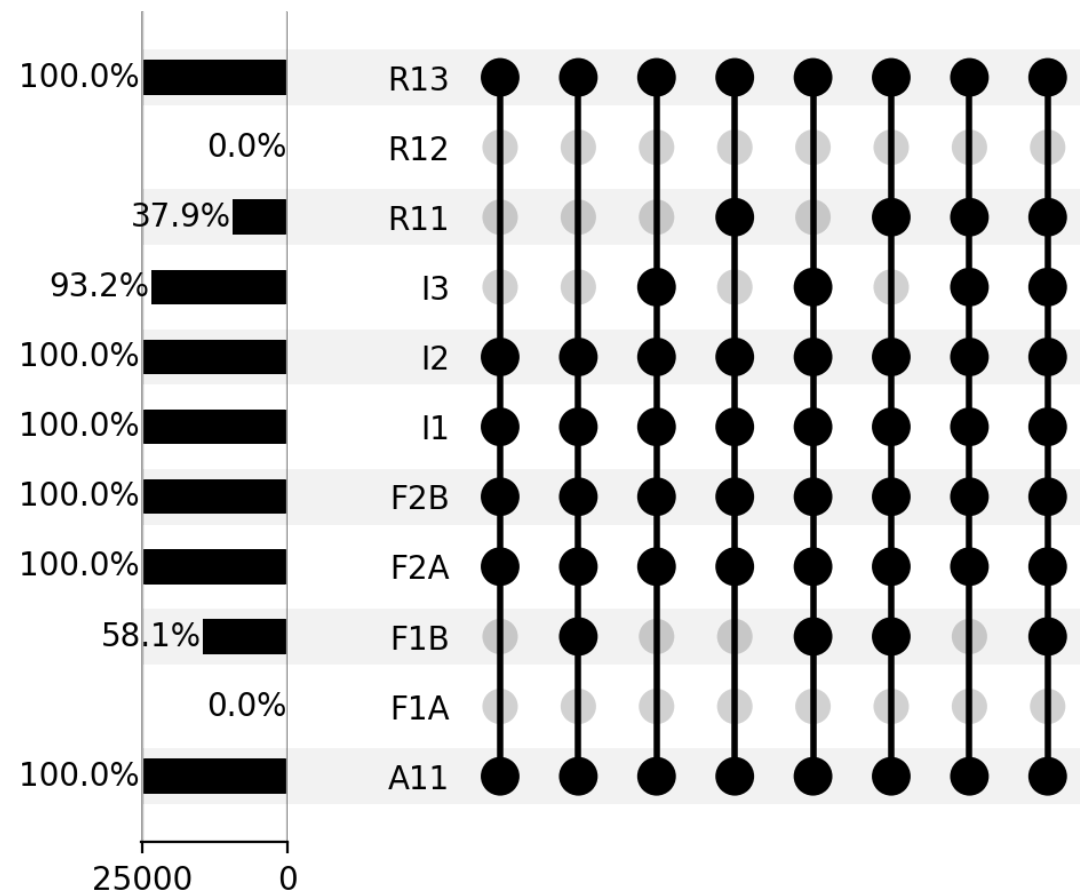
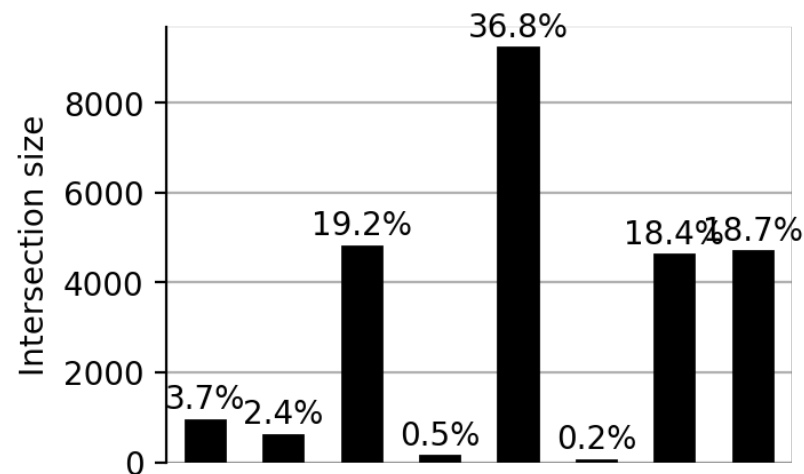
### Properties

|                                                                               |     |     |           |
|-------------------------------------------------------------------------------|-----|-----|-----------|
| <a href="http://ogp.me/ns#description">http://ogp.me/ns#description</a>       | OLS | LOV | BioPortal |
| <a href="http://ogp.me/ns#site_name">http://ogp.me/ns#site_name</a>           | OLS | LOV | BioPortal |
| <a href="http://ogp.me/ns#title">http://ogp.me/ns#title</a>                   | OLS | LOV | BioPortal |
| <a href="http://ogp.me/ns#url">http://ogp.me/ns#url</a>                       | OLS | LOV | BioPortal |
| <a href="http://schema.org/affiliation">http://schema.org/affiliation</a>     | OLS | LOV | BioPortal |
| <a href="http://schema.org/author">http://schema.org/author</a>               | OLS | LOV | BioPortal |
| <a href="http://schema.org/contentSize">http://schema.org/contentSize</a>     | OLS | LOV | BioPortal |
| <a href="http://schema.org/contentUrl">http://schema.org/contentUrl</a>       | OLS | LOV | BioPortal |
| <a href="http://schema.org/creator">http://schema.org/creator</a>             | OLS | LOV | BioPortal |
| <a href="http://schema.org/dateCreated">http://schema.org/dateCreated</a>     | OLS | LOV | BioPortal |
| <a href="http://schema.org/dateModified">http://schema.org/dateModified</a>   | OLS | LOV | BioPortal |
| <a href="http://schema.org/datePublished">http://schema.org/datePublished</a> | OLS | LOV | BioPortal |

Using FAIR-Checker at  
large scale

# Large-scale FAIR metrics evaluations

How FAIR are Bio.Tools registered softwares ?

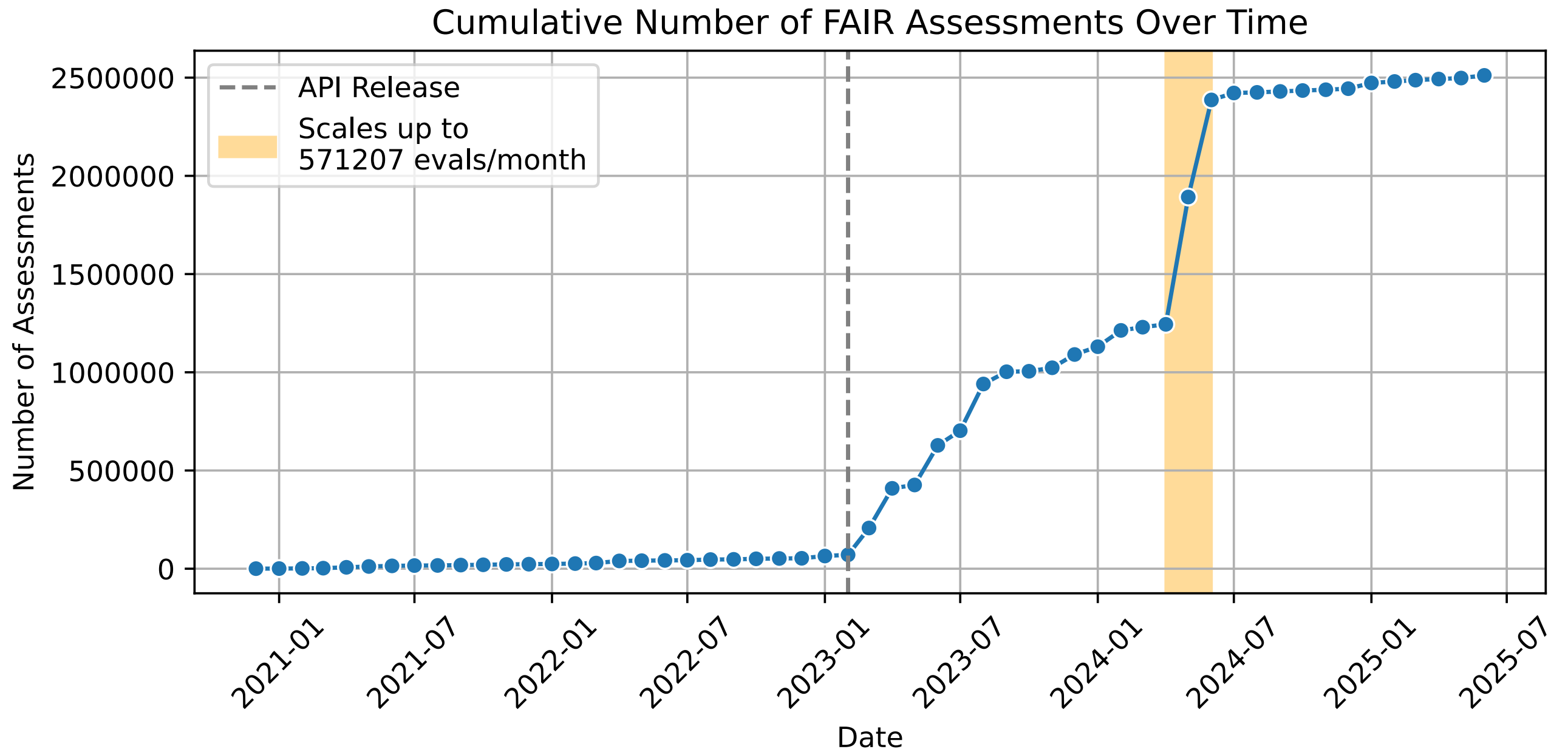


Running FAIR-Checker over  
**25.000+** bioinformatics softwares  
from Bio.tools.

R1.1: Only 37,9% of the tools  
expose a **licence**

R1.2: **No provenance** metadata  
→ massive impact if bio.tools  
developers provide PROV / PAV  
ontology terms

# Usage statistics

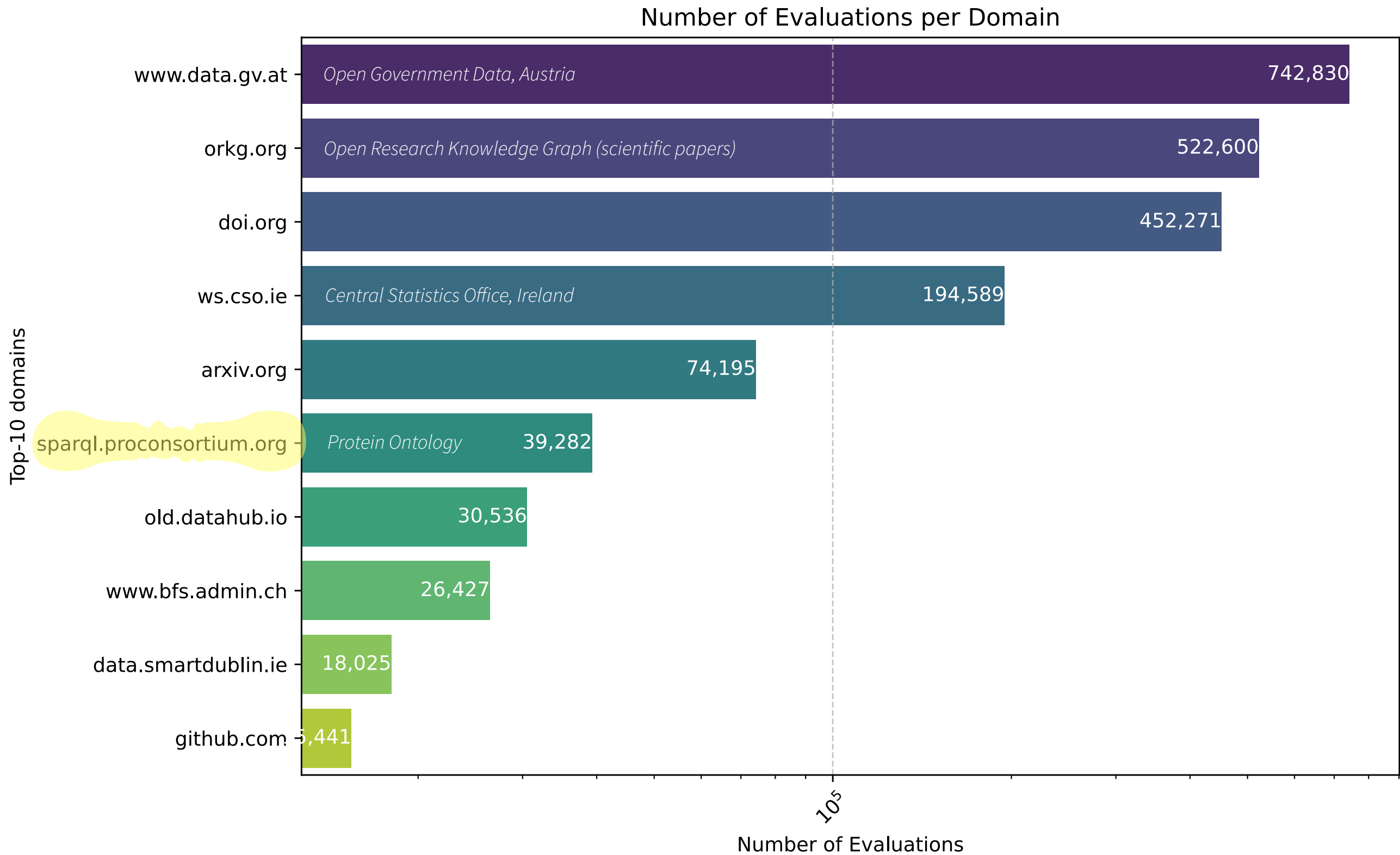


**102,180** unique target URLs

**47,537** unique target URLs evaluated more than 2 times

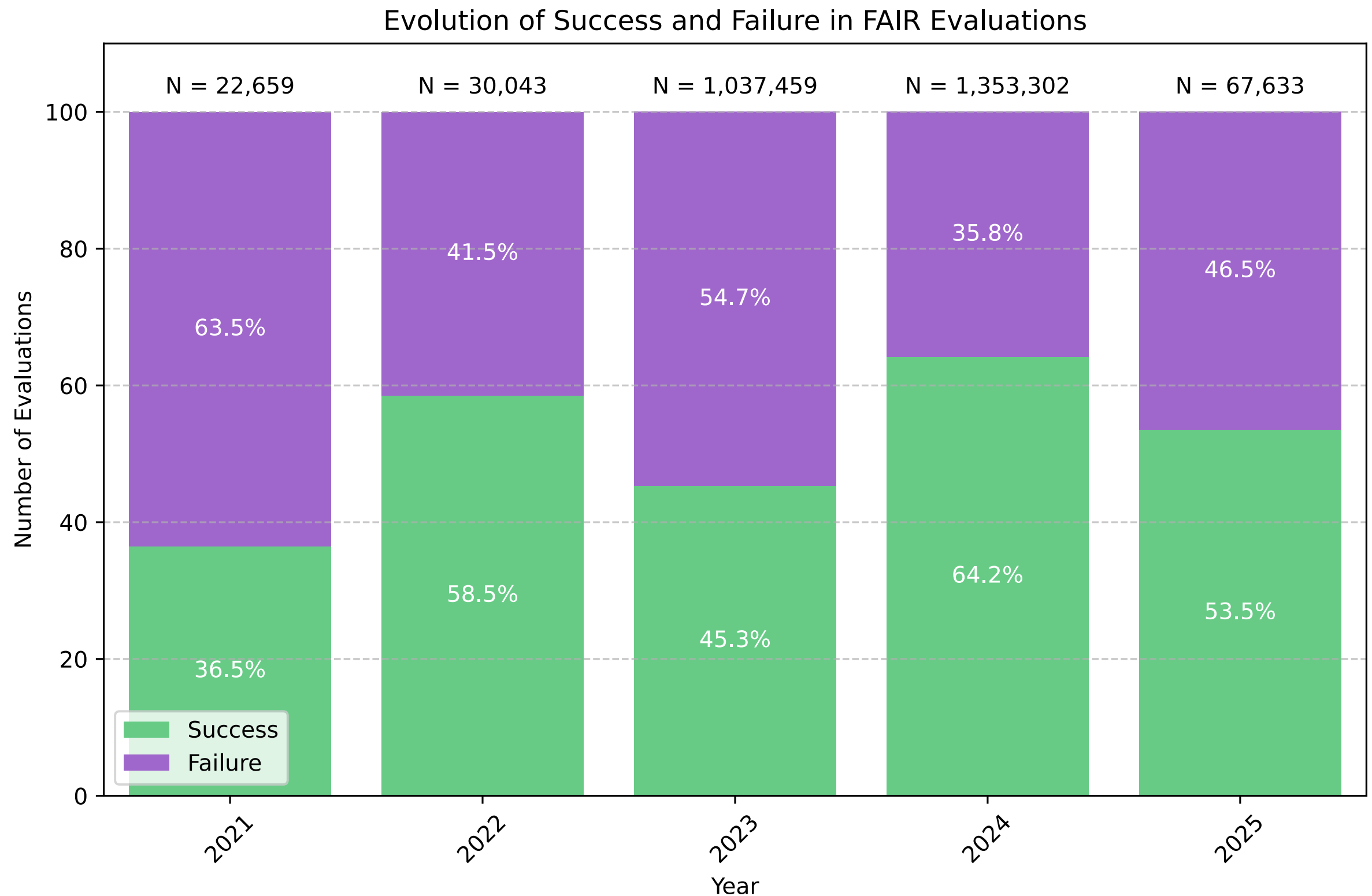
**437** unique target URLs evaluated more than 10 times

# Most evaluated domains



# Evolution of FAIR scores

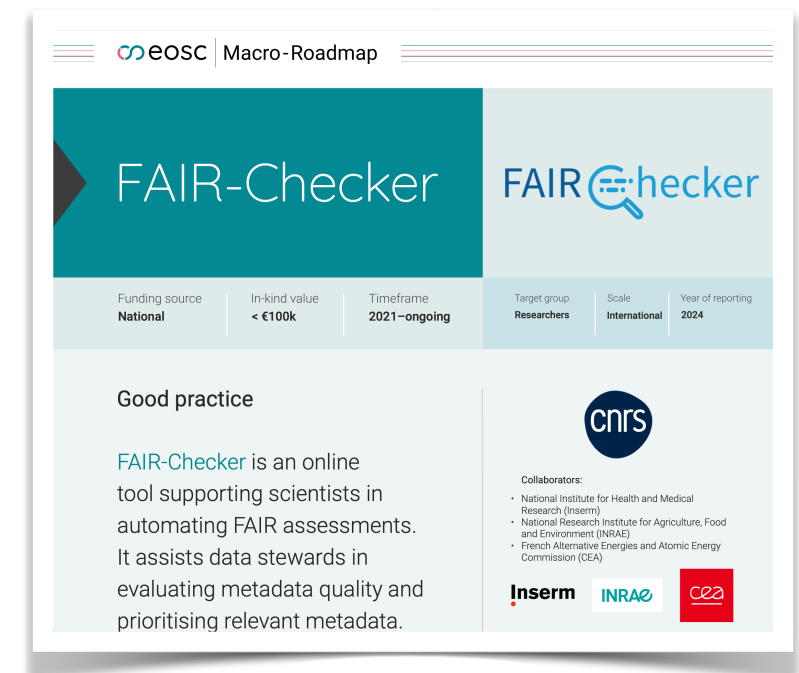
Work in progress



# A community service, with possible improvements

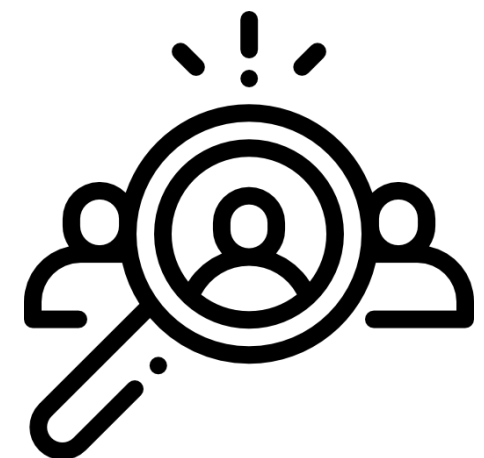


<https://fair-checker.france-bioinformatique.fr>



## Future works

- ▶ Support "FAIR-Signposting" for **better metadata consumption**
- ▶ Bioschemas **profile recommender**
- ▶ Allow users to provide **missing metadata**
- ▶ Suggest semantic metadata based on **AI generation** pipeline
- ▶ Retrospective **usage study**





Jennien Andersen



Sylvie Cazalens



Philippe Lamare

## A Generic Framework to Better Understand and Compare FAIRness Measures

Philippe Lamarre<sup>1</sup>, Jennie Andersen<sup>1</sup>, Alban Gaignard<sup>2</sup>, and Sylvie Cazalens<sup>1</sup>

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`firstname.lastname@insa-lyon.fr`

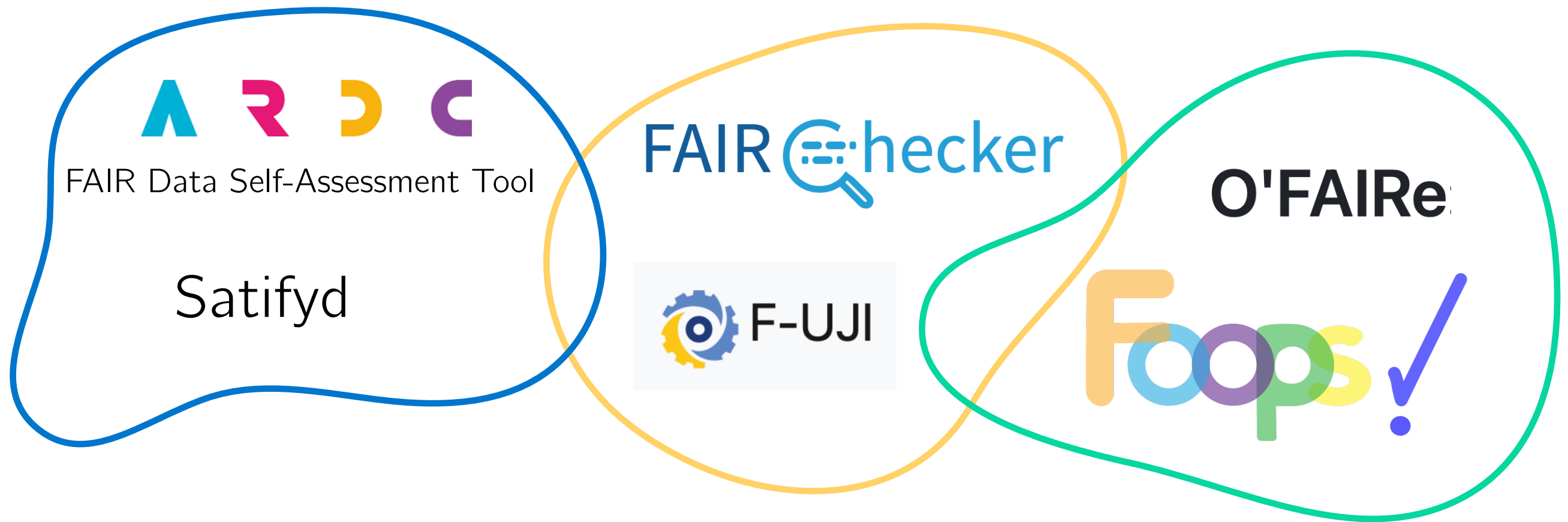
<sup>2</sup> Nantes Université, CNRS, INSERM, l'institut du thorax, F-44000 Nantes, France  
`alban.gaignard@univ-nantes.fr`

**Abstract.** In recent years, the adoption of the FAIR principles has achieved notable success. This progress has led to the development of numerous assessment tools originating from diverse fields of application, thus addressing diverse object types, interpretations and implementations. Given the plethora of proposals available, it is crucial for users to precisely understand these measures, compare them effectively, make

# Many FAIR assessment tools ...

# → do they provide similar results ?

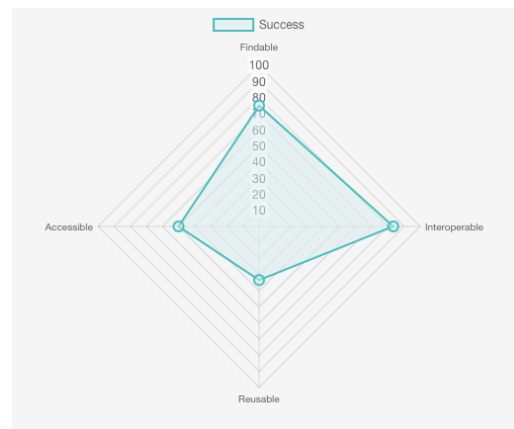
# Multiple FAIR assessment tools



- ▶ Check lists / questionnaires
- ▶ Automated tools
- ▶ Some tools are "community oriented"
- ▶ Some tools are "technologically opinionated"

# How FAIR is a bioinformatics software ?

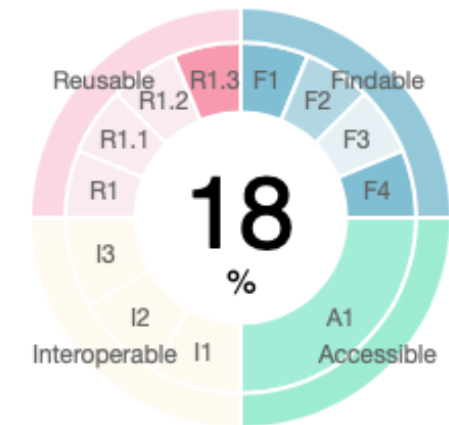
FAIR  checker



FAIR assessment 62.50 %



<https://bio.tools/seurat>



|                | Score earned: |                                                                                       | Fair level: |
|----------------|---------------|---------------------------------------------------------------------------------------|-------------|
| Findable:      | 2.5 of 7      |  | initial     |
| Accessible:    | 1 of 3        |  | initial     |
| Interoperable: | 0 of 4        |  | incomplete  |
| Reusable:      | 1 of 10       |  | initial     |

- ▶ Why are results so different?
- ▶ Which tool should I use?
- ▶ What's behind the scene?

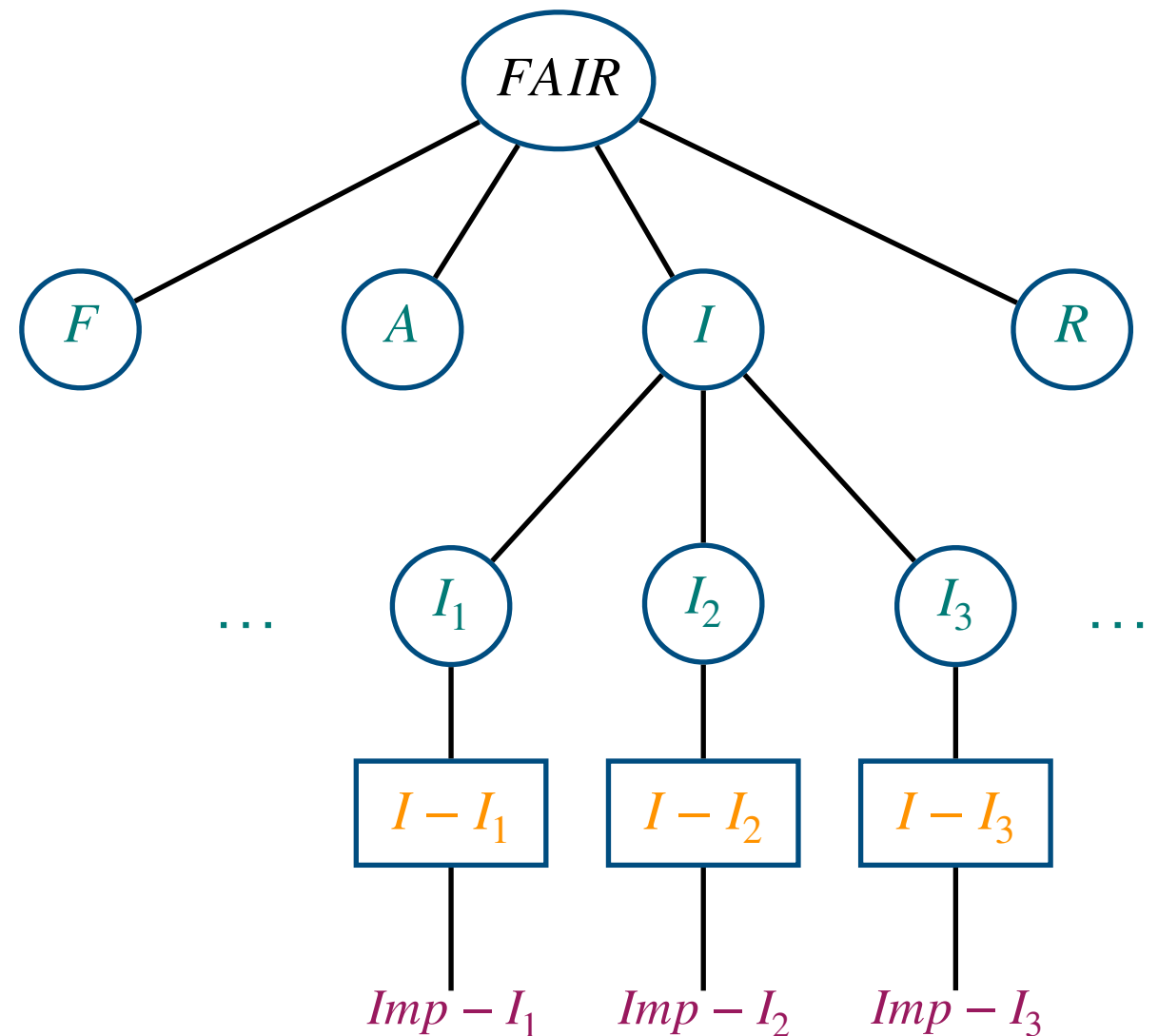
① We need a computable  
framework for *modeling*  
FAIR measures

# Objectives

- (i) **specify measures** in a uniform and computable **model**
- (ii) propose specific **quantities** to **analyse and compare** FAIR assessment approaches

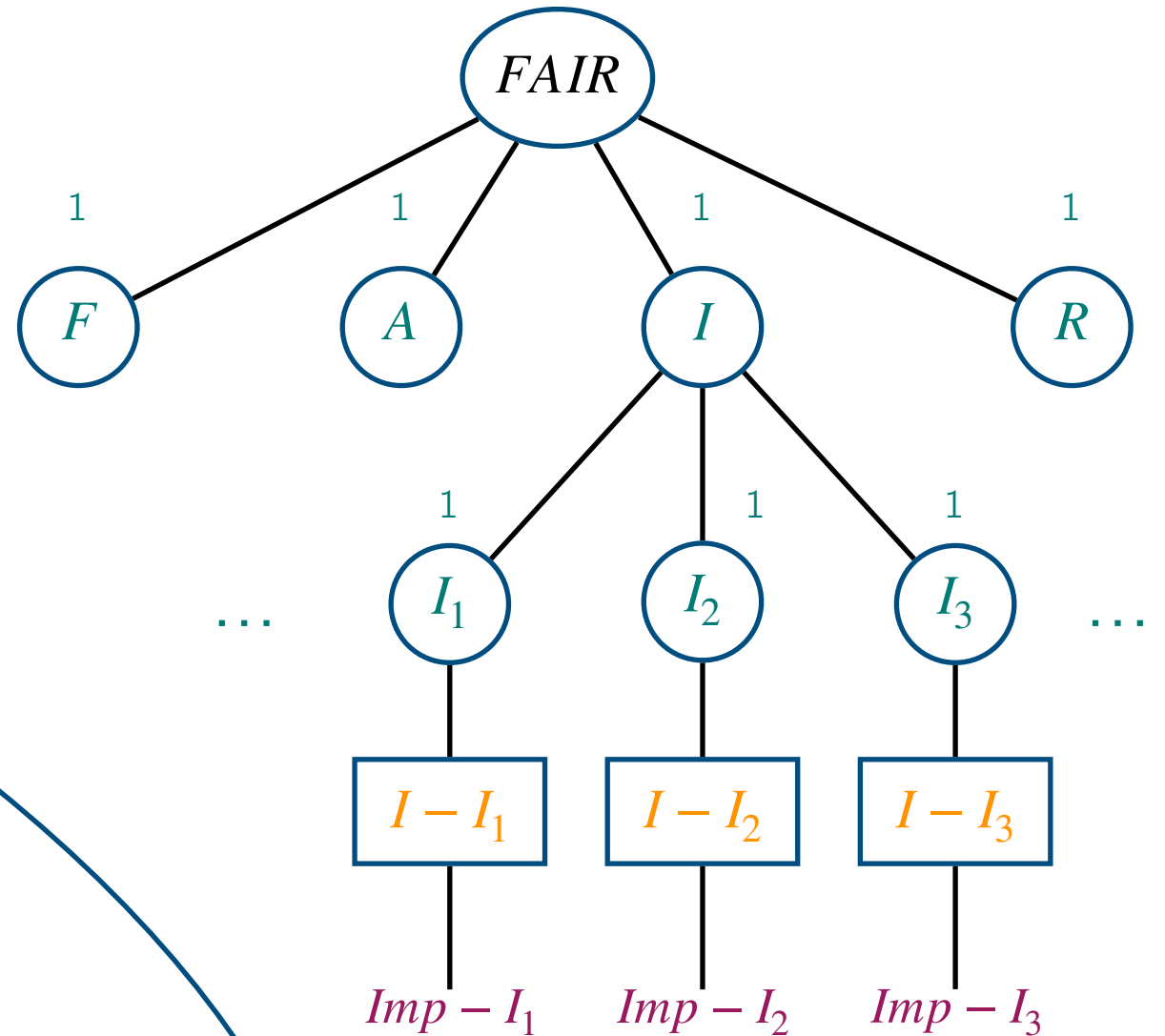
# Typical structure for FAIR assessment methods

- Principles  $P$ : the FAIR principles and their sub-principles
- Indicators  $I(\mathcal{M})$ : the specification of principles, (i.e. what has to be verified)
- Implementations  $Imp(\mathcal{M})$ : the implementations of the principles



$$\mathcal{M} = (\overbrace{V, E, FAIR}^{\text{structure}}, \underbrace{\diamond, w, v_{max}}_{\text{score}}, D)$$

# Score computation



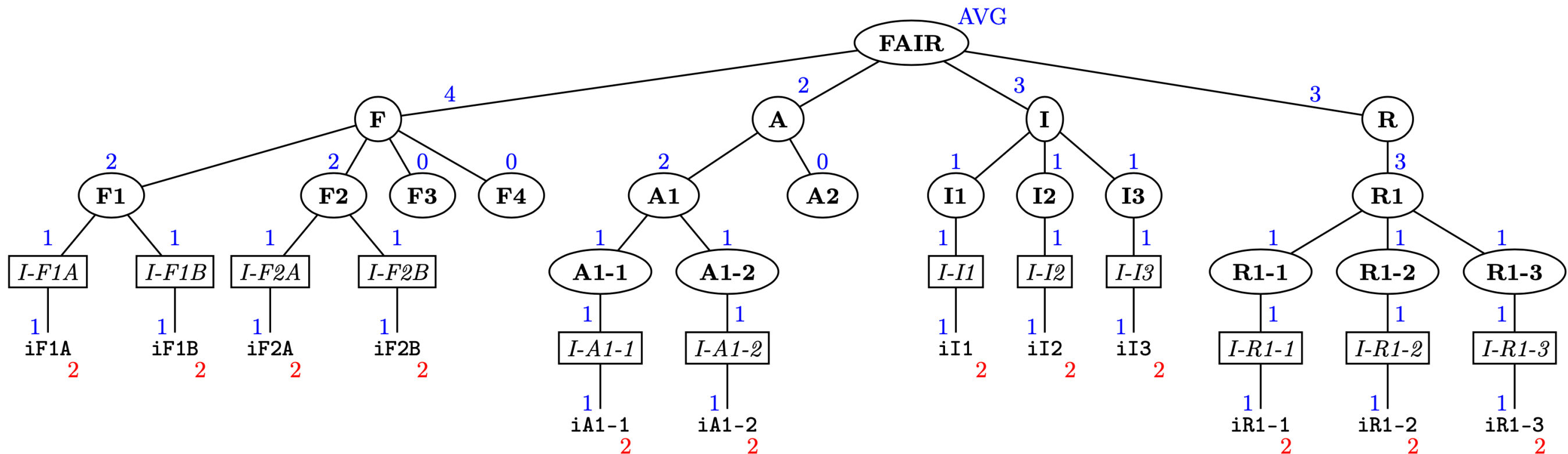
Max function:  $v_{max}(i)$  returns the maximum reachable score for an indicator or an implementation

Weighting function:  $w(n)$  returns the weight of the node  $n$  compared to its siblings

Aggregation function : weighted sum or weighted average to collect scores

$$\mathcal{M} = (\overbrace{V, E, FAIR}^{\text{structure}}, \underbrace{\diamond, w, v_{max}, D}_{\text{score}})$$

# Model instantiation with FAIR-Checker

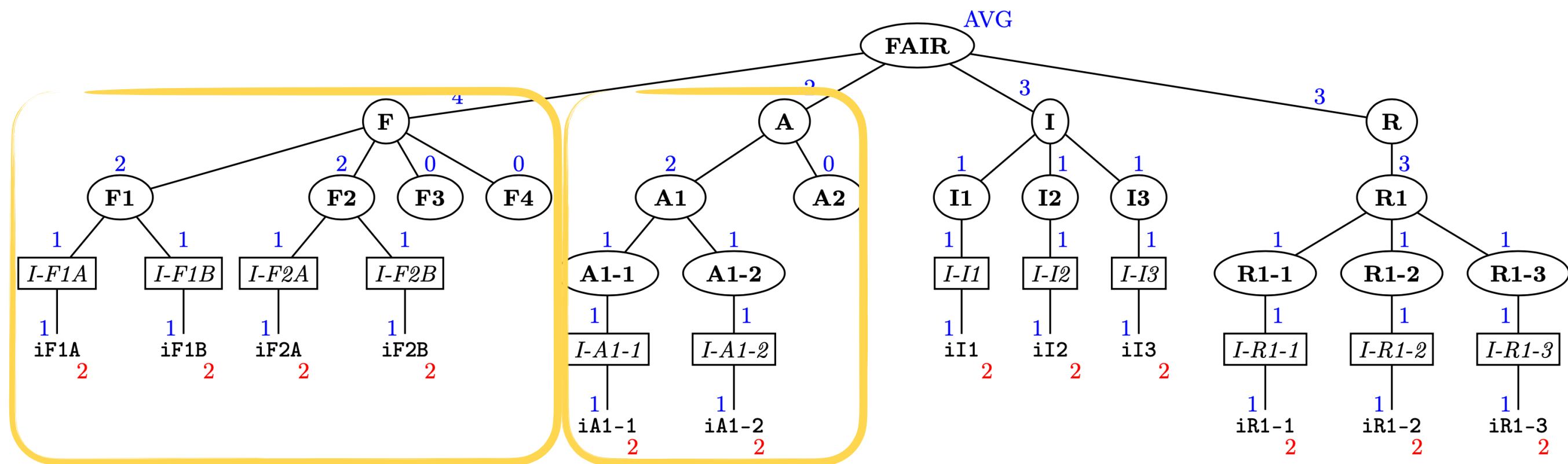


- ▶ The tool fits the tree-based structure  $\{P, I, Imp\}$
- ▶ needs the maximum reachable score per implementation  $\rightarrow 2$
- ▶ needs an aggregation function  $\rightarrow AVG$  (*normalized scores*)
- ▶ needs cumulative weights for each principle  $\rightarrow w \in [0,4]$

② We need quantities to  
characterize and compare  
FAIR measures

# Quantifying the *granularity* of a measure

**Idea:** a global metric computed as the mean number of indicators per principles with at least one indicator. **If we have one principle with many implementations → important granularity.**

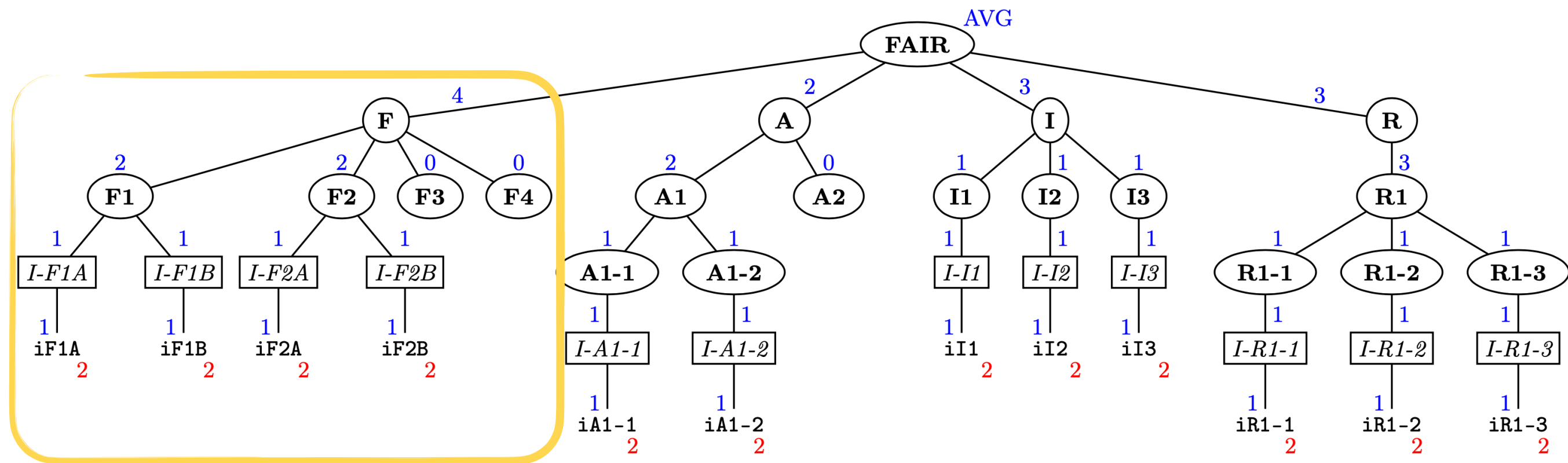


F has 4 indicators, A has only 2 → principles have not the same granularity

$$\text{granularity}(F) = 4/2 = 2 ; \text{granularity}(\text{FAIR}) = 12/10 = 1.2$$

# Quantifying the *coverage* of a measure

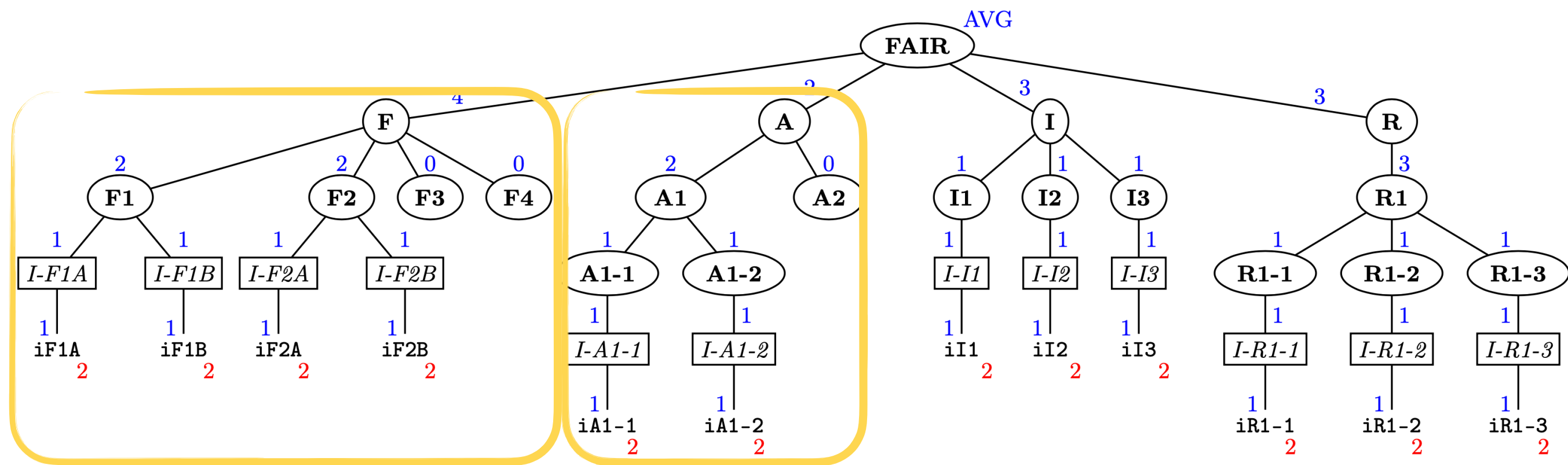
**Idea:** a principle is considered as covered if it exists at least one implementation for the principle or, its sub-principles.



F3 & F4 are never evaluated → F has only 50% of coverage

# Quantifying the *impact* of principles

**Idea:** the impact of a principle corresponds to the **global score** obtained when **all** its underlying **implementations are successful** (without considering other principles).



$$\text{imp}(F) = 4 \cdot 2 / 12 \cdot 2 = 2/6$$

$$\text{imp}(A) = 2 \cdot 2 / 12 \cdot 2 = 1/6$$

→ F contributes 2 times more compared to A in the global FAIR score

# Experimental results

# Experimental setup

- ▶ Evaluated tools

automated: F-UJI, FAIR Evaluator, FOOPS!, O'FAIRe, FAIR-Checker,  
questionnaires: ARDC, SATiFYD

- ▶ 10 selected web resources for FAIR assessments, covering

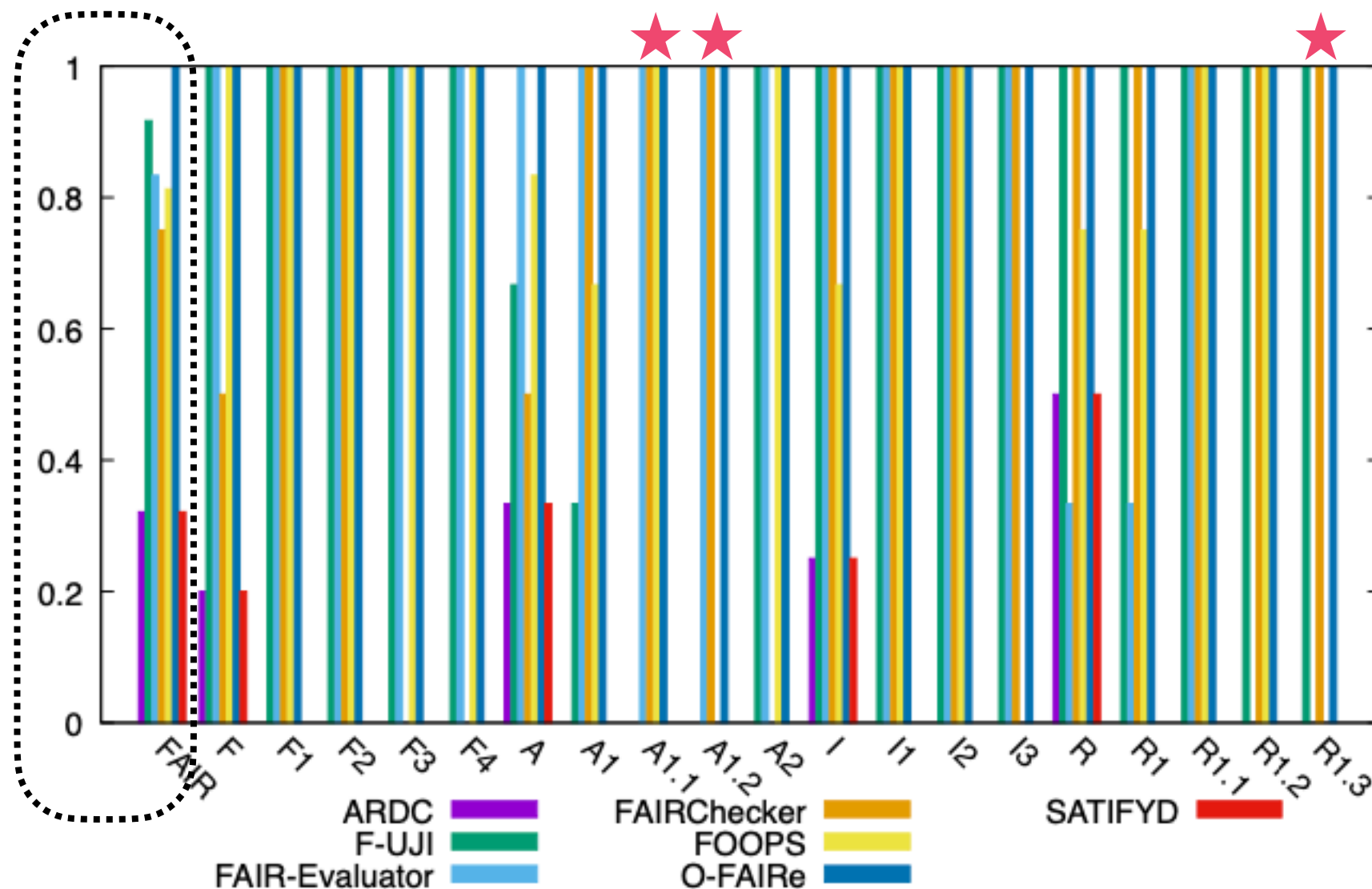
- datasets descriptions
- ontologies
- online courses
- bioinformatics software
- RDF files

# Do engines reach consensus on FAIR assessment ?

| Resource                                   | F-UJI (%) |                                                                                      | FAIR-Checker (%) | Std dev |
|--------------------------------------------|-----------|--------------------------------------------------------------------------------------|------------------|---------|
| Dataset (PANGAEA) [31]                     | 91        |                                                                                      | 91.70            | 0.49    |
| Gene Ontology (OLS) [21]                   | 18        |   | 16.70            | 0.92    |
| Dataset (Harvard Dataverse) [23]           | 75        |                                                                                      | 79.20            | 2.97    |
| Dataset (Kaggle) [26]                      | 60        |                                                                                      | 70.80            | 7.64    |
| Online course (Moodle) [28]                | 4         |                                                                                      | 16.70            | 8.98    |
| ..... Dataset (Governmental platform) [22] | 52        |                                                                                      | 70.80            | 13.29   |
| Dataset (WHO) [39]                         | 27        |  | 50.00            | 16.26   |
| Training material (TeSS) [36]              | 39        |                                                                                      | 70.80            | 22.49   |
| Bioinformatics tool (bio.tools) [6]        | 18        |                                                                                      | 54.20            | 25.60   |
| Dataset (RDF metadata) [33]                | 43        |                                                                                      | 87.50            | 31.47   |

- ▶ Higher scores for FAIR-Checker
- ▶ Last two entries: std. dev. > 25 % ?

# Coverage rates

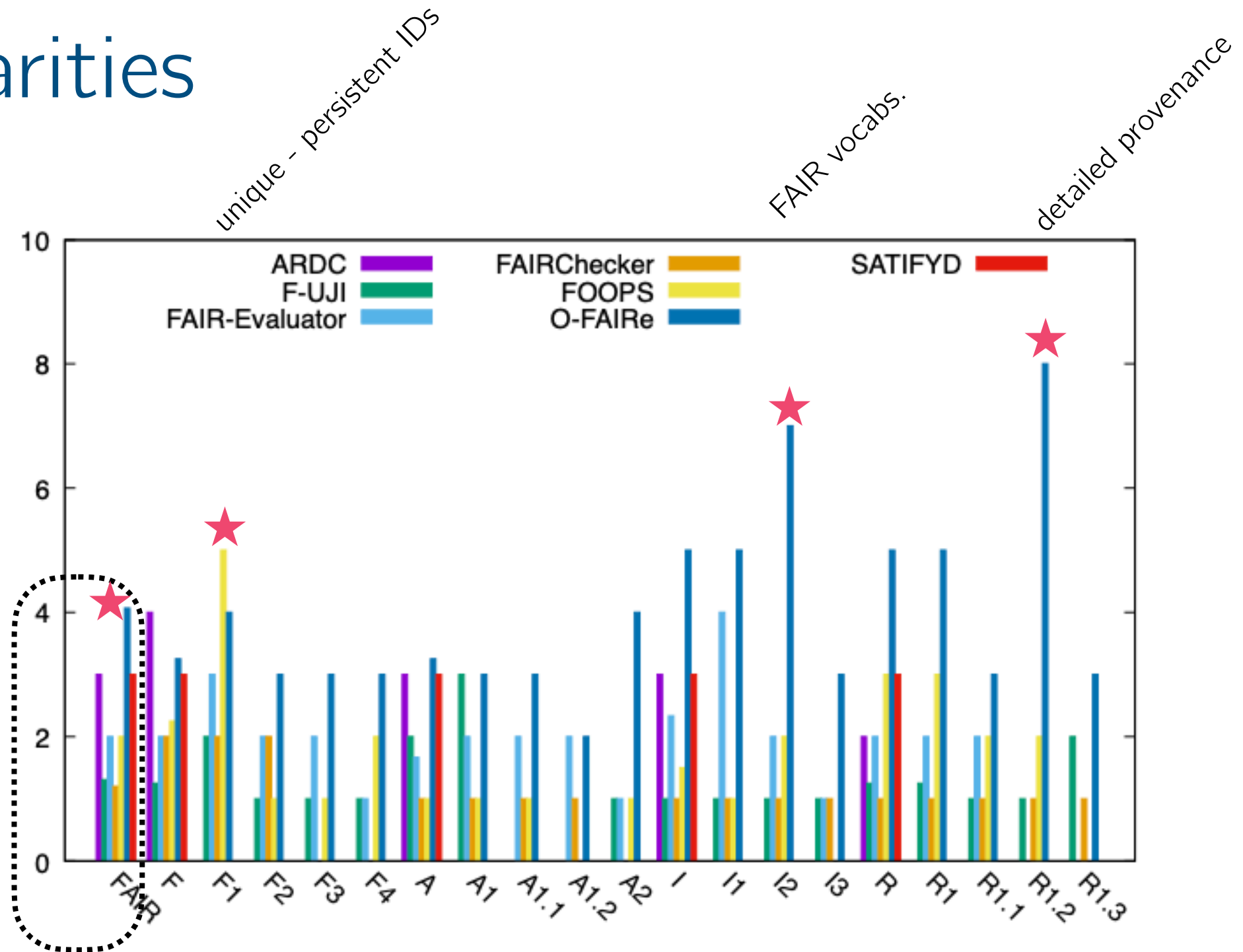


some principles (★) are not implemented  
 → technical issues ?  
 → interpretation ?

{O-FAIRe} → the only one with a full coverage (1)

Low coverage for questionnaires (no sub-principles)

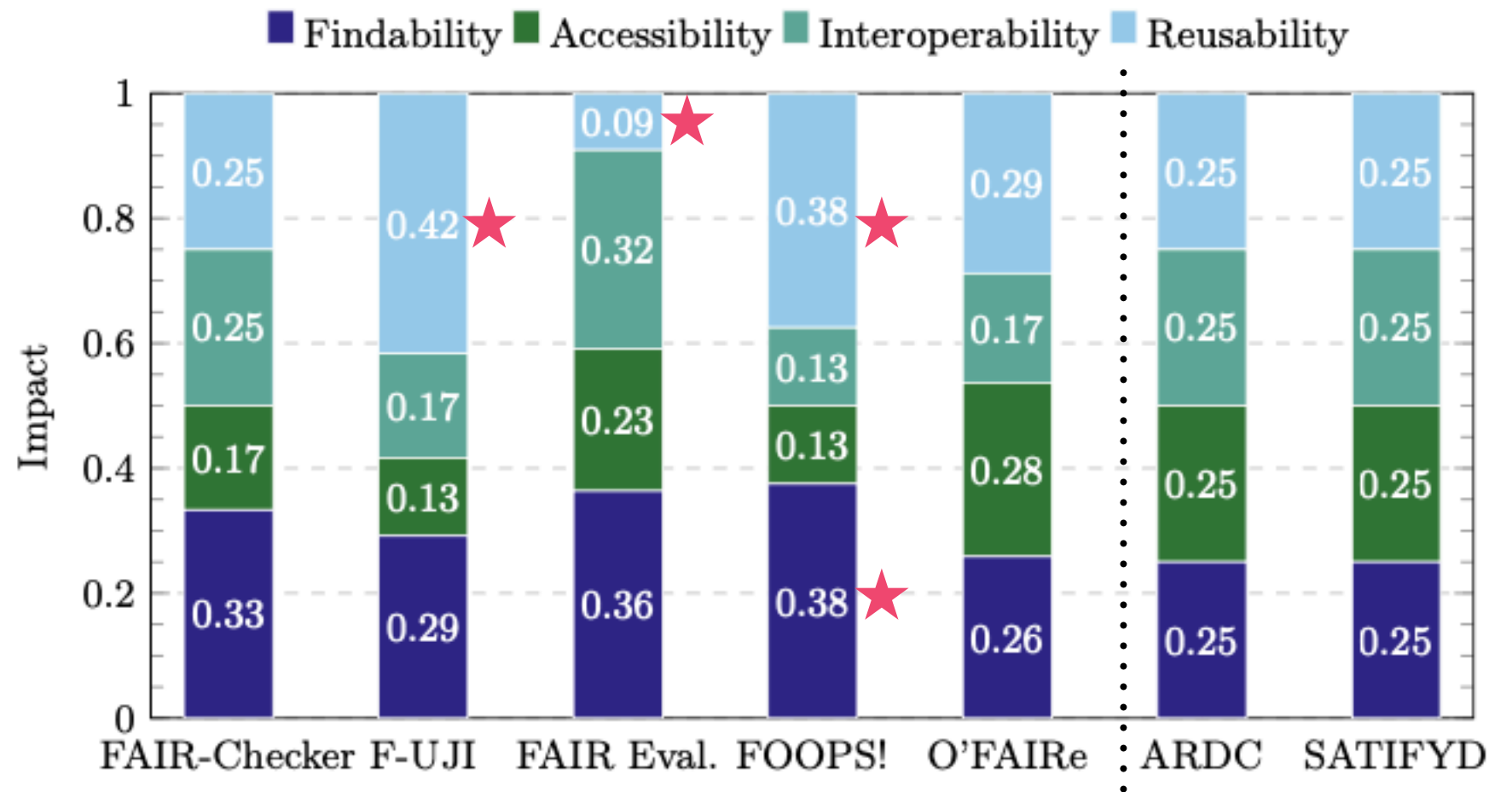
# Granularities



- ▶ O-FAIRe (targeting ontologies) has the most fine-grained evaluation (e.g. I2 and R1.2) → great diversity in metadata for ontologies
- ▶ FOOPS deeply investigate F1 (identifiers)

# Impacts

*Are all principle equally contributing to the global FAIR assesment score ?*



→ *How to get a good FAIR score with a minimal effort ?*

- ▶ Pay attention to identifiers (F), license + provenance + domain-specific standards (R) if you use FOOPS!
- ▶ Not useful to spend energy on provenance or domain ontologies if you use FAIR-Evaluator ...
- ▶ ... but pay attention to it if you use F-UJI.

# Conclusion and future works

# Take-home message

- ▶ We need to understand the specificities of FAIR measurement tools
- ▶ We introduced a generic **model** for representing FAIR measures and computing their **granularity**, **coverage** and **impact**
- ▶ Some **tools are "biased"** (intentionally or not):
  - they explore more deeply some dimensions
  - which has an impact on the scores
- ▶ **Future works** include
  - better investigating **implementations** (e.g. dependencies), with tools developers
  - **share** these metadata on the web (e.g. DQV ontology)
  - contribute to **FAIR harmonization efforts** with other communities (softwares, workflows, machine learning resources ...)

# Acknowledgements



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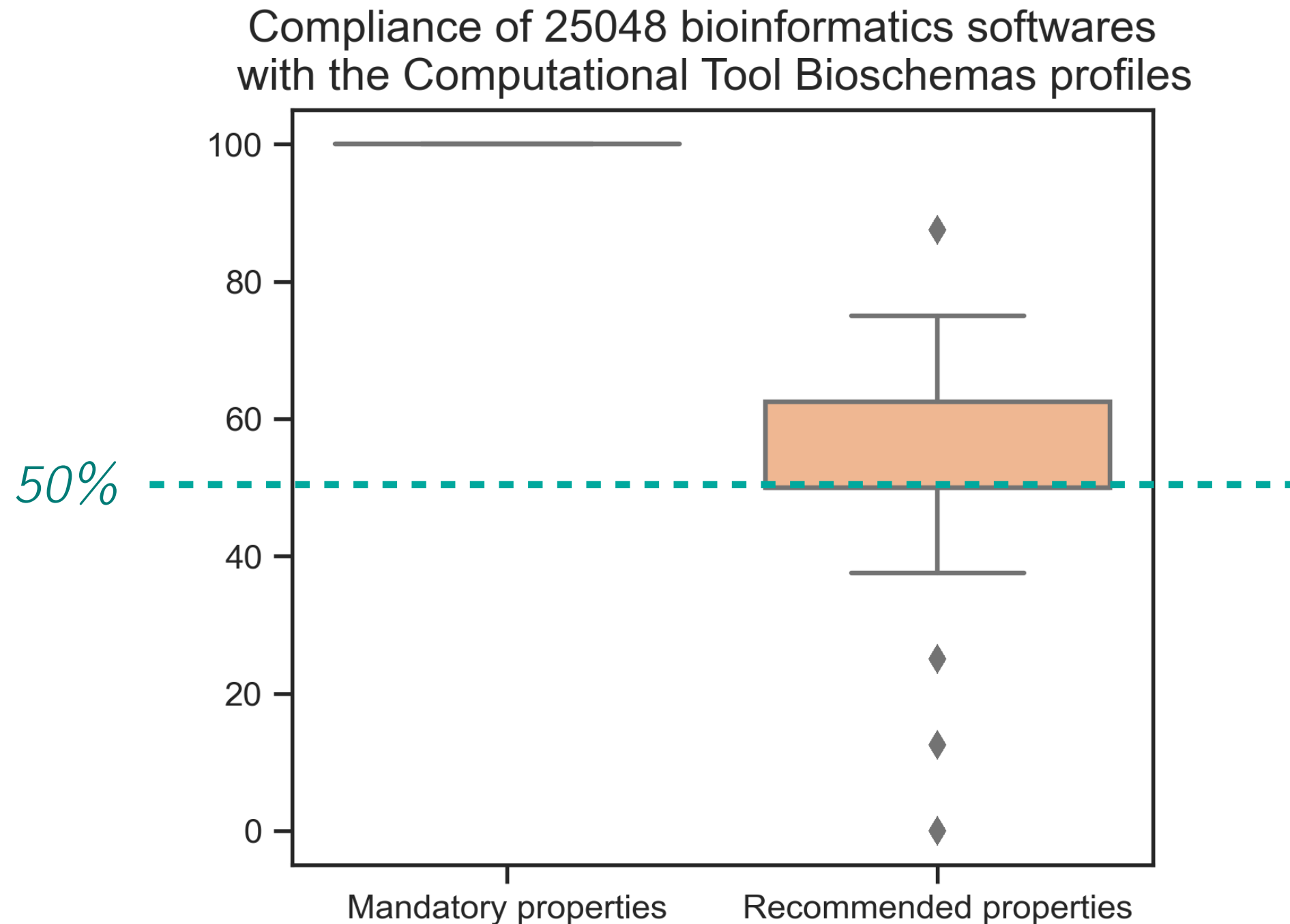
contact

[alban.gaignard@univ-nantes.fr](mailto:alban.gaignard@univ-nantes.fr)

Back-up slides

# SHACL shapes evaluations

→ machine-actionable bioschemas profile



# Impact

$$best_n(imp) = \begin{cases} v_{max}(imp) & \text{if } imp \in \text{desc}(n) \\ 0 & \text{otherwise} \end{cases}$$

$$\text{impact}(\mathcal{M}, n) = \frac{\text{score}(\mathcal{M}, best_n, \text{FAIR})}{\text{score}(\mathcal{M}, best_{FAIR}, \text{FAIR})}$$

# Coverage

$$\text{cover}(\mathcal{M}, n) = \begin{cases} 0 & \text{if } \text{children}_{\mathcal{M}}(n) = \emptyset \\ \frac{\left( \sum_{n' \in \text{children}_{\mathcal{M}}(n) \cap P} \text{cover}(\mathcal{M}, n') \right) + \text{local}(\mathcal{M}, n)}{|\text{children}_{\mathcal{M}}(n) \cap P| + \text{local}(\mathcal{M}, n)} & \text{else} \end{cases}$$

# Granularity

$$\text{gran}(\mathcal{M}) = \text{gran}(\mathcal{M}, \text{FAIR}) \quad (5)$$

where, for  $n \in P$ :

$$\text{gran}(\mathcal{M}, n) = \frac{|\text{desc}_{\mathcal{M}}(n) \cap I(\mathcal{M})|}{|\{p \in (\{n\} \cup \text{desc}_{\mathcal{M}}(n)) \cap P, \text{children}_{\mathcal{M}}(p) \cap I(\mathcal{M}) \neq \emptyset\}|} \quad (6)$$